

Information Discrete Mathematics & Calculus

A Foundation from the Retained Difference — axioms, definitions, operators, theorems, principles, and a complete discrete calculus

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Core Epistemic Structure

- **Core Respondent:** Yaoharee Lahtee — lived experience, selections, and interpretation behind

the readout-first stance and every design choice in this repository.

- **Interactional Expert:** None.
- **AI Model(s) Used:** Claude Fable 5.1 (via Claude Code) — Toledo code registration,

documentation of codes.

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***Edition note (v1.0.0 — release; completeness-first, proofs next).** This release organises the existing knowledge base and re-synthesises it from the latest findings into one coherent foundation, so it can be **used to do physics calculations precisely** in the readout-first framework. It prioritises **complete coverage** — from the roots up through a full information discrete calculus, the powerful analytic tools, and the reconciliations with the external record. **Proofs are stated by reference to the machine-checked witnesses where they exist and are otherwise [results-first] sketches deferred to the next version (v-proofs).** The goal is the closed, usable skeleton, not yet the full proof tower.*

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- **Part 0** — Foundations of the foundation (the commitment; injected infinities/zeros)
- **Part 0.5** — The reader, resolution, and coarse-graining (the substrate: A , G_λ , admissible $E = \tilde{E} \circ G_\lambda$, sets/functions/equality, S-layers, A8)
- **Part I** — Retained-Distinction Logic (RDL)
- **Part II** — The primitive δ_R and the genesis of number (RD1–RD9, D)
- **Part III** — The number tower $\mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R}$ (the continuum as a readout)
- **Part IV** — Geometry from retained difference (distance = resistance; angle = overlap)
- **Part V** — The retained-information operator L_R ($B(\Phi, \Phi) = I(\Phi)$)
- **Part VI** — The continuum as a readout; the injected-infinity diagnosis
- **Part VII** — Complete operator reference ($\neg \forall \rightarrow \leftrightarrow \forall \exists$, $= \neq < \leq$, $+ - \times \div ^ \sqrt{} \log$); §7.0 the information language
- **Part VIII** — Information discrete calculus (D_ε , I_ε , FTCC, series, spectral window)
- **Part IX** — Powerful tools re-adapted (Ramanujan summation, Euler–Maclaurin, zeta reg., q-calculus, ...)

- **Part X** — Integrated extensions (pairs/products/functions from δ_R ; cardinality; analysis; probability; geometry-from-the-graph; and reconciliations with causal-set theory, network/spectral-graph theory, Bishop, category theory, Shannon)
- **Part XI** — Closing the continuum (discrete derivative/integral/limit/ODE/special-function; the closure theorem)
- **Part XII** — Abstract algebra as retained symmetry (group = $\text{Aut}(F, 0)$; ring/field; Lagrange; Galois)
- **Part XIII** — Linear algebra as the algebra of L_R ($\mathbb{Q}$ -vector space; metric; determinant; exact solve)
- **Part XIV** — Complex analysis as the rotation-readout of retained phase (i = quarter-turn; roots of unity; holomorphy; residue)
- **Part XV** — Combinatorics & graph theory (the native home of δ_R ; L_R connectivity; matrix-tree; Euler χ)
- **Part XVI** — Measure theory & functional analysis (discrete measure; L^p ; finite spectral theorem; Riesz)
- **Part XVII** — Category theory in the readout vocabulary (G_λ reflector; fibration; setoids; finite Yoneda)
- **Part XVIII** — Statistics & inference (retained-frequency estimation; a test IS a verdict; Bayesian reweighting)
- **Part XIX** — Optimization (gradient = D_ε ; convexity; Lagrange; exact- $\mathbb{Q}$ linear programming)
- **Part XX** — The continuum-maya bridge (the continuum constructed as a readout, Λ , and computed with; exact FTCC core)
- **Part XXI** — The frontier without the continuum: paradox dissolution (topology · manifolds · PDE — the decisive stance)
- **Part XXI.5** — The Hilbert-space mathematical core (finite-dim exact; operators, spectral, tensor/partial-trace, CP/PPT; the $+\mathbb{R}$ fence)
- **Part XXII** — The root design: three axioms that generate the solver (READOUT · FOLD · DECISION + the KEYSTONE bridge; the 230-kind branch map — 259 live with the Hilbert core and the polynomial_positivity certificate solver; reductions machine-checked in `IDM_Reduction.v`)
- **Appendices A/B/C** — contaminated-concept table · machine-checked theorem index · axiom-dependence discipline
- **Appendices D/E** — validation: 1000 problems ประถม $\rightarrow$ ปริญญาเอก (1000/1000) · 100 continuous problems reproduced from the discrete (100/100)

Using this to calculate physics (the release purpose)

To compute a physical quantity in this framework: (1) declare the **resolution** λ / ε and the **admissible readout** $E = \tilde{E} \circ G_\lambda$ (Part 0.5); (2) express the quantity in the **operator language** (Part VII §7.0) and the **causal finite- ε calculus** (Part VIII: D_ε , I_ε , FTCC) —

never a continuum primitive; (3) read spectra / mass-ratios / distances from `L_R` and Born-rule **overlap fractions**, not from angles or coordinates (Parts IV–V, VII); (4) if a divergent or asymptotic expression appears, use the **re-adapted tools of Part IX** (regularization = the finite retained readout), declaring the limit only after an **A8 stability** proof; (5) tier every result (`Th_coqc / finite_diagnostic / Dr / +R-Open`) and refuse the continuum endpoints. The pre-write checklist of the companion skill enforces this.

A world-class mathematical foundation built from ONE primitive — the retained difference — and carried, by definition-and-theorem, up to the number tower, discrete analysis, geometry, and the operator of information, with the continuum recovered as a readout of the discrete root. Every statement carries an honest tier. The machine-checked core is axiom-free over $\mathbb{Q}$; where the continuum enters, it is flagged, never smuggled.

How to read this book — the tier discipline (never collapse)

Every numbered result is tagged:

- **Th_coqc** — machine-checked in Coq, **axiom-free over $\mathbb{Q}$** (`Print Assumptions` = `*Closed under`

the global context*; a non-vacuous witness). Named witnesses point to the verified corpus.

- **Th_coqc+classic** — machine-checked, resting only on the single isolated axiom `classic`

(excluded middle) in the classical layer.

- **+R-axioms** — machine-checked but importing `Coq.Reals` (the 2 real-analysis axioms); NOT

axiom-free — a *readout* rung, honestly flagged.

- **Th_coqc-elig** / **finite_diagnostic** — a finite, decidable computation or proof *eligible* for

Coq (all objects finite lists over decidable-equal $\mathbb{Q}$) but not necessarily checked yet; or a numeric readout reproduced to a **declared tolerance**. The working tag from Part VIII onward.

- **Dr** — a stance / definition / interpretation over checked pieces (design-narrative).
- **+R-Open** (a.k.a. **Open**) — genuinely unsettled, or needs the completed continuum (`+reals`).

Stated and predicted as a readout, never smuggled in as a premise.

The load-bearing rule. Never assert *all/every/universal* over a finite witness set without tagging `Dr`. Never put a continuum name on a finite analogue. Never claim to *solve* a continuum question — diagnose it as a non-readout and predict the readout. The boundary between the provable and the Open is the boundary of the infinity axioms.

Notation

`δ_R` primitive retained difference · `D` the naturals-as-semiring · `ℤ, ℚ, ℝ` the number tower · `<` retained order · `L_R` the graph Laplacian (retained-information operator) · `{·, ·}_G` the retained inner product · `Th` theorem, `Def` definition, `Ax` axiom, `Pr` principle · `Verdict(ACCEPT/HOLD/BLOCK)` the fail-closed result type and `eq_eps/solve_obstruction` the numeric-honesty discipline of `tools/idm_discipline.py` (truthy only on `ACCEPT`; used from Part XVIII on).

On the `Th_coqc` witnesses (warrant disclosure). Most `Th_coqc` citations of a bare witness file (`RD.v`, `RDL.v`, ...) point to the **external machine-checked corpora** — `solver-arc-private`, `readout_genesis`, `readout_universe` — where those `coqc`-verified proofs live (Appendix B/C). In addition, this repository now ships its **own** axiom-free witnesses in `formal/IDM_FiniteWitnesses.v` (Coq 8.20; `coqc` exit 0, `Print Assumptions` = Closed under the global context for every result) — currently the operator keystone `Th 5.1 B(Φ, Φ)=I(Φ)` (`IDM_Keystone.v`), plus `Th 10.1` (Kuratowski pair injectivity), §15.2 handshake, §15.3 pigeonhole, §17.2 finite-Yoneda, §12.2 semiring distributivity (`IDM_FiniteWitnesses.v`). Where no witness exists yet, the tag is `Th_coqc-elig` (eligible, not yet checked), never a bare `Th_coqc`.

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Part 0 — Foundations of the foundation

0.1 The one commitment (Pr 0.1, Dr)

Everything an agency ever reads is a finite retained difference — a readout, rational and discrete. The continuum ($\mathbb{R}$), infinite divisibility, and $+\infty$ are non-readouts: real as boundaries, never as appearances.

This is a stance, not a proof that classical mathematics is wrong; a classical realist may hold the continuum ontologically actual. We adopt the commitment and follow it to the end. Its mechanical witness: the entire core is **Print Assumptions**-Closed *precisely because* it refuses the infinity axioms (**Require Reals** is where they would enter).

0.2 The injected infinities and zeros (Def 0.2)

An "open/hard/paradoxical" problem is, until shown otherwise, an **artifact of a silently injected non-readout**. Name it before deferring:

- **I1** $\mathbb{R}$ -completeness (LUB/Dedekind) · **I2** infinite divisibility $h \rightarrow 0$ · **I3** infinite scale

separation $\text{Re}, \wedge \rightarrow \infty$ · **I4** actual $+\infty$.

- **Z1** the point $r=0$ · **Z2** reached continuum $h=0$ · **Z3** absolute rest $v=0, T=0$ · **Z4**

the true void.

- **Pr 0.2 (reciprocity, Dr)**. $1/0=\infty$: zero and infinity are one non-readout seen from two

sides; they appear together at every singularity. A finite reader lives strictly *between* the refused endpoints and touches neither. *A refused endpoint being unreadable is the system working, not a wall to break through.*

0.3 What a number, a unit, a symbol, and an equation ARE (a philosophy of mathematics)

Every mathematical object admits **two levels of account, and is never only one** — a distinction of pure philosophy of mathematics (designation vs constitution; token vs type), not of any doctrine.

- **The conventional (designative) level** — what the object is *as it appears to a reader*:
a

readout $r=0(X)$ at a declared resolution λ , an admissible designation $E=\tilde{E} \circ G_{\lambda}$.

- **The ultimate (constitutive) level** — what it is *in its root constitution*: ****retained**

distinction $\delta_R = (a \# b)$ and the structure L_R it induces. Meaning is generated downstream of the retained trace; a domain name is a translation q_D (with $q_D \circ F = F \# \circ q_D$), never a new root** [epistemic-foundation synthesis, Lahtee 2026].

Read through both levels:

- A **number** is neither a Platonic object nor a bare mark. It is a **readout-invariant of a retained-distinction structure**: the numeral *designates*, the equivalence class $[X]$ under reader-equivalence ($X \sim X' \Leftrightarrow 0(X) = 0(X')$) constitutes. (π, φ are the worked case — §4.4.)

- A **unit** is not physical stuff; it is a **calibration/gauge on readouts** — a choice of scale on

the retained metric $\langle \cdot, \cdot \rangle_G$. Changing units is a reader-relabelling that preserves every readout: a redundancy, not a fact.

- A **symbol** is not the thing; it is a **stable reader-label** for an equivalence class. One class

may bear many symbols; use $\neq$ mention, token $\neq$ type.

- An **equation** $A=B$ is not a metaphysical identity of objects; it is the **admissible indistinguishability** of two constructions to every reader ($0(A) = 0(B)$) at a declared resolution — the operational $=$ of §0.5.3 / §7.0.

So the old dispute is sidestepped (Dr — this framework's reading, not a claim to have settled the philosophy-of-mathematics literature). The number is not an abstract object floating free of any reader (against Platonism), nor an empty inscription (against strict formalism), nor merely a useful fiction (against fictionalism): it is the **invariant of a reading** — conventional in its appearance, constitutive in its retained structure, and neither without the other. **Governing principle:** never import a mathematical object as a primitive premise; *earn* it as a downstream readout of retained distinction, tier-tagged.

Part 0.5 — The reader, resolution, and coarse-graining (the substrate)

Before number or logic can be *used*, we fix **who reads** and **at what resolution**. This Part supplies the collection / function / equality substrate — *grounded in coarse-graining, not assumed* — and the finite-resolution discipline that lets analysis proceed without ever injecting the continuum. (*Integrates the author's earlier **Causal Calculus: Primitive Aggregation under Finite Causal Access** — Lahtee 2026, DOI 10.5281/zenodo.18164015 — whose still-usable core is extracted here.*)

0.5.1 Access & coarse-graining (Def 0.5.1)

$\mathbf{IR}$ — the world, **never read directly**. Access map $\mathbf{A} : \mathbf{IR} \rightarrow \mathbf{I_acc}$ to the accessible information states. Coarse-grain family $\mathbf{G}_\lambda : \mathbf{I_acc} \rightarrow \mathbf{I}_\lambda$, λ a resolution. A reader sees only $\mathbf{I}_\lambda$ — what survives coarse-graining at its resolution.

0.5.2 Admissible description (Def 0.5.2 — the formal readout)

$\mathbf{E}$ is **admissible at resolution λ** iff there is $\tilde{\mathbf{E}}$ with $\mathbf{E} = \tilde{\mathbf{E}} \circ \mathbf{G}_\lambda$ — the result does NOT distinguish what $\mathbf{G}_\lambda$ discards. **Consequence:** answers are free of any $\varepsilon \rightarrow 0$ limit; sub-resolution differences are *not perceived*. This is `readout-not-truth` made operational: a reading is admissible exactly when it **factors through the coarse-graining**.

0.5.3 The collection / function / equality substrate (induced by $\mathbf{G}_\lambda$)

- **collections / sets** := the fibers of $\mathbf{G}_\lambda$ (a "set" is a class of $\mathbf{I_acc}$ -states identified at resolution λ).
- **functions / maps** := **admissible maps** — those that factor through $\mathbf{G}_\lambda$.
- **equivalence** $\sim_\lambda := \mathbf{G}_\lambda(x) = \mathbf{G}_\lambda(y)$ (indistinguishable at λ); ****ordered pairs / products /**

quotients** are the corresponding admissible constructions.

- **equality** $=$:= mutual indistinguishability to every reader — the finest $\sim_\lambda$ — now

non-circular: grounded in the coarse-grain fibers, not defined using $=$ itself (this repairs Def 7.2). *This is the sets-and-functions layer a foundation needs, **induced by coarse-graining rather than posited**.*

0.5.4 Causal order (S2)

reference $\mathbf{s}_0$; $\mathbf{t(s)} := \min \# \mathbf{steps(s_0 \rightarrow s)}$ — the minimal-step metric (= distance as retained resistance, Def 4.1); $\mathbf{s \leq s' \Leftrightarrow t(s) \leq t(s')}$; the **causal index** $\mathbf{n := t(s_n)}$ generates the tape; resolutions $\varepsilon_t, \varepsilon_x > 0$.

0.5.5 The layer & infinity discipline (the tier system, operationalized)

- Layers: **S1** (pre-theoretic — descriptions, resolution, admissibility) · **S2** (causal order) · **S3+** (constructed operators / equations at declared finite resolution).

- **A11** — declare the layer of every result. **A8 (infinity discipline)** — *work at finite ε first; take $\varepsilon \rightarrow 0$ (or $N \rightarrow \infty$) ONLY after a stability proof, and declare it.* **CMG** — effective, resolution-dependent scope; no fundamental claims. **A8 is the analysis discipline that lets the whole calculus be built without ever making an actual infinity primitive.**

0.5.6 Honest metatheory note (declared, not hidden)

This substrate still uses ordinary **finitary logic** (propositional connectives, quantifiers, induction on indices) as its meta-language, and the machine-checked corpus is verified in **Coq's type theory (CIC)**. "One primitive" means the **philosophical** primitive (δ_R — retained distinction under finite access); the *metallogic* in which its theorems are proved is declared here, not smuggled. The refusal of the continuum is a refusal to make it a **primitive appearance**, not a denial that a finitary metatheory is in use.

Part I — Retained-Distinction Logic (RDL)

The logic must tolerate a local contradiction without collapse, because a reader may hold conflicting retained records before resolution.

1.1 Axioms of RDL (Ax-RDL, paraconsistent)

- **Ax-RDL1 (distinction).** The basic judgement is a *distinction* $a \triangleright b$ (a is told apart from b),

not a truth value.

- **Ax-RDL2 (contradiction = obstruction).** A contradiction is a *nonzero obstruction* $\perp_o$, an

object, not a licence to derive everything.

- **Ax-RDL3 (non-explosion).** From $p \wedge \neg p$ one may NOT derive an arbitrary q (*ex falso* is

rejected).

- **Ax-RDL4 (gap $\neq$ assertion).** Absence of a distinction (a gap) is not the assertion of its

negation.

- **Ax-RDL5 (classical limit).** When every obstruction is zero, RDL reduces to classical logic.

1.2 Theorems of RDL (Th_coqc, axiom-free — witness formal/RDL.v, 8 theorems)

- **Th 1.1 (soundness).** RDL's calculus is sound for its retained-distinction semantics.
- **Th 1.2 (non-explosion).** $p \wedge \neg p \vdash \bot$ for arbitrary p . (*the core paraconsistency guarantee*)
- **Th 1.3 (contradiction = obstruction).** A contradiction denotes a nonzero obstruction object.
- **Th 1.4 (gap $\neq$ assertion).** A truth-value gap is provably distinct from an assertion.
- **Th 1.5 (classical limit).** Zero-obstruction RDL $\equiv$ classical propositional logic.
- **Th 1.6 (non-triviality).** RDL has a model with a genuine (non-zero) obstruction — it is not

vacuous.

Part II — The primitive and the genesis of number

2.1 The primitive (Def 2.1, `Dr` → `Th_coqc` realization) [Toledo: `delta_R`]

`δR` — *a retained difference exists*. Ontologically primitive; nothing is assumed below it. Its realization is machine-checked: **Th 2.1** (`primordial_difference_exists` [Toledo: `delta_R/M.03.v1`], `Th_coqc`) — a first distinction exists, witnessed by `succ 0 ≠ 0`.

2.2 Axioms of Retained Difference (Ax-RD1...RD9) [Toledo: `RD1`, `RD2`, `RD3`, `RD4`, `RD5`, `RD6`, `RD7`, `RD8`, `RD9`]

The engine `D` is generated by:

- **RD1** there is a ground `0` (no prior distinction).
- **RD2** a successor `succ` (retain one more distinction).
- **RD3** `succ n ≠ 0` (a retained tick is never the ground). (*witness* `RD3_succ_ne_zero`)
- **RD4** `succ` is injective (distinct histories stay distinct).
- **RD5** induction: a property holding at `0` and closed under `succ` holds throughout.
- **RD6–RD9** the recursion clauses defining `⊕` (retained addition) and `⊗` (retained composition)

and the order `<` from `succ`.

2.3 The naturals `D` [Toledo: `D`] (`Th_coqc`, axiom-free — witness `RD.v`, `RDL_Distinguishability.v` in the

sister repo `solver-arc-private/formal/`, NOT local to this repo's own `formal/`; verified directly — `coqc -q RD.v` then `Print Assumptions <name>` reports *Closed under the global context* for every theorem below except `Con_PA_classical`, see Th 2.7)

- **Th 2.2 (semiring).** $(D, \oplus, \otimes, 0, 1)$ is a commutative semiring.
- **Th 2.3 (total order + well-ordering + strong induction).** $(D, <)$ is a total order, well-ordered.
- **Th 2.4 (isomorphism).** `toNat` : $D \rightarrow \mathbb{N}$ is a semiring-and-order isomorphism $D \cong \mathbb{N}$.
- **Th 2.5 (elementary equivalence).** $D \equiv \mathbb{N}$ via a full first-order satisfaction transfer

(witness `sat_transfer` / `sentence_transfer`). **Tier note (see §10.3's downgrade):** the Coq proof itself is machine-checked axiom-free, but §10.3 flags the *statement* as `+R-Open` on philosophical grounds (it quantifies over a completed model of `D`) rather than on proof-gap grounds. Both are stated here rather than silently picking one — reconciling which tier governs a Coq-checked but completed-totality-quantified statement is an open call for this repo's author, not settled by re-running the proof.

- **Th 2.6 (D models PA; Con_PA).** $D \models$ first-order Peano Arithmetic (witness D_models_PA),

and **PA is consistent** $Con_PA : \neg Prov\ PA \perp$ (consistency-from-a-model), axiom-free. The Con_PA half is unconditionally Th_coqc by any reading (it is the non-collapse of D 's own structure, not a claim about a completed model). The $D \models PA$ half carries the same tier note as Th 2.5 above.

- **Th 2.7 (Con_PA_classical, Th_coqc+classic).** The classical frontier of consistency is closed

on the single axiom `classic`. A parameterized variant that removes even this axiom exists — `solver-arc-private/formal/RD_ConPA_ReadoutBivalence.v: Con_PA_classical_param` — verified directly: `Print Assumptions Con_PA_classical_param` reports *Closed under the global context* (no `classic`), via an explicit, disclosed `ReadoutComplete` hypothesis instead of a global axiom.

- **Th 2.8 (second-order categoricity).** Dedekind categoricity of the second-order theory.
- **Th 2.9 (discrete floor, Th_coqc).** $\neg \exists z, 0 < z < succ\ 0$ — nothing lies below the first tick.

Principle (Pr 2.1): *density / the continuum is provably ABSENT at the root*; the continuum can only ever be a later readout.

Part III — The number tower (each rung a readout of the last)

3.1 The integers $\mathbb{Z}$ [Toledo: Z] (Th_coqc, axiom-free — Grothendieck completion of $\mathbb{D}$)

- **Def 3.1.** $\mathbb{Z} := (\mathbb{D} \times \mathbb{D}) / \sim$ with $(a, b) \sim (c, d) \Leftrightarrow a \oplus d = c \oplus b$; value homomorphism to the standard $\mathbb{Z}$.
- **Th 3.1.** $(\mathbb{Z}, +, \cdot, 0, 1)$ is a commutative **ring** (zadd_neg additive inverse, zmul_assoc,

zmul_distrib_l, all up to zeq).

3.2 Discrete calculus over $\mathbb{Z}$ (Th_coqc, axiom-free — *calculus with no reals*)

- **Def 3.2.** difference $\Delta f(n) := f(n+1) \ominus f(n)$; sum Σ .
- **Th 3.2 (discrete FTC, both directions).** FTC, FTC_inverse (telescoping): Σ and Δ are

mutually inverse.

- **Th 3.3 (discrete Leibniz).** $\Delta(f \cdot g) = f \cdot \Delta g + (\Sigma g) \cdot \Delta f$ (product rule), no continuum needed.

3.3 The rationals $\mathbb{Q}$ [Toledo: Q] (Th_coqc, axiom-free — field of fractions of $\mathbb{Z}$)

- **Def 3.3.** $\mathbb{Q} := (\mathbb{Z} \times \mathbb{Z}_{\neq 0}) / \sim$; value homomorphism into the standard $\mathbb{Q}$.
- **Th 3.4.** $(\mathbb{Q}, +, \cdot)$ is a **field**: qadd_neg, qmul_assoc, qmul_distrib_l, and the

multiplicative inverse qmul_inv. The ladder $\mathbb{D}$ semiring $\rightarrow \mathbb{Z}$ ring $\rightarrow \mathbb{Q}$ field is verified end-to-end (69 constructive theorems, axiom-free).

3.4 The reals $\mathbb{R}$ [Toledo: R] — the continuum AS a readout (Th_coqc, axiom-free unless noted; Toledo tiers this rung +R-axioms, not Th_coqc, since it imports Coq.Reals)

- **Def 3.4.** $\mathbb{R} :=$ Bishop **regular Cauchy sequences of $\mathbb{Q}$** : $f: \mathbb{N}^+ \rightarrow \mathbb{Q}$ with

$|f\ n - f\ m| \leq 1/n + 1/m$. Equality Req (an equivalence). Injection inj_Q : $\mathbb{Q} \hookrightarrow \mathbb{R}$.

- **Th 3.5 (ordered field up to Req).** $\mathbb{R}$ is a commutative ring, ordered (Rle partial order,

antisymmetric; strict Rlt irreflexive/transitive), **multiplication** (Rmul well-defined, associative, distributive) and **reciprocals for every real apart from 0** (Rinv_gen , Rmul_inv_gen , either sign). (the ladder $D \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R}$ closed at the field level, axiom-free)

- **Th 3.6 (Cauchy-completeness).** R_complete / R_complete_metric — every real Cauchy sequence

has an **explicitly constructed** limit (the diagonal $L_n := X(2n)(6n)$): *the continuum point is read off the discrete rational approximants. This is the thesis of the book, machine-checked.*

- **Th 3.7 (constructive order).** Rlt is **cotransitive** (Rlt_cotrans) — the constructive substitute for trichotomy.
- **Th 3.8 (lattice).** binary Rmax / Rmin with universal properties — the constructive finite

suprema/infima.

- **Pr 3.1 (honest non-constructivity, explicitly marked).** Full **trichotomy**, a **total $\leq$** , and the **classical least-upper-bound property** are each **NOT constructively valid** — they imply an omniscience principle (LPO/WLPO). Cotransitivity + Cauchy-completeness + the finite lattice are their correct constructive replacements. (To assume LUB is to inject $I1$.)

- **Th 3.9 ($+\mathbb{R}$ -axioms where continuum analysis is used).** ε - δ continuity Rcontinuous_at , the

metric-space laws (Rdist_triangle), Lipschitz operations — machine-checked as *derived rungs*; the full derivative/integral tower and continuum geometry remain **imported** (Open / $+\text{reals}$), honestly deferred.

Principle (Pr 3.2, the tower). $\delta_{\mathbb{R}} \rightarrow D \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R}$: each rung is *defined from* the previous; the continuum is the last rung and is a **readout** of the discrete root, never its substrate. π , e , φ are **readout-invariants** (Appendix A), diagnostics that reconstruction succeeded — only their finite $\mathbb{Q}$ -approximants ever appear (Th A.1: $\text{InfoIrrationalNonReadout}$ — $\sqrt{2}$ is a non-readout).

Part IV — Geometry from retained difference

4.1 Distance (Def 4.1, Th_coqc core)

Distance enters not as a coordinate difference but as **accumulated retained resistance along the optimal path**: $\text{dist}(x,y) := \min \text{ over paths } x \rightsquigarrow y \text{ of } \sum (\text{edge resistances})$.

- **Th 4.1 (metric-space laws).** $\text{dist}(x,x)=0$, symmetry, non-negativity, and the **triangle**

inequality — the last *because a detour cannot be cheaper than the direct optimum*.
(dist_* , L^1 product metric dist2_* .)

- **Th 4.2 (Tarski betweenness).** $\text{Betw}(x,y,z)$ axiomatized and proved on the discrete metric — the

synthetic-geometry primitive, with no real coordinates.

4.2 Angle without the continuum (Def 4.2, Th_coqc) — *the contamination this book most guards*

The textbook **angle** ($\text{acos} / \text{atan2}$, degrees) requires $\mathbb{R}$ -completeness = I1 and is refused. In its place:

- **Def 4.2a (overlap fraction).** for retained states v, e :

$\text{overlap}(v,e) := |\langle v,e \rangle_G|^2 / ((\langle v,v \rangle_G \cdot \langle e,e \rangle_G))$ — a **Born-rule ratio**, rational (+, ·, ÷ only), no trig, no π , no $\mathbb{R}$. This is the quantity $\cos^2 \theta$ would have given, arrived at without ever injecting the continuum.

- **Def 4.2b (turning number).** a rotation is a **rational fraction of one full cycle**, not a real

angle. Mixing/holonomy read out as overlap fractions or turning numbers.

4.3 The relation graph and the operator L_R [Toledo: L_R] (Th_coqc, axiom-free — RDL_CausalOrder.v)

- **Def 4.3.** the retained relation graph; the interval count $N(x,y) = \text{toNat } y - \text{toNat } x - 1$

($N_{\text{count_eq}}$), $N=0 \Leftrightarrow$ no event between.

- **Th 4.3 (Laplacian mechanics).** $L_R (= \text{lap} / B)$ is **symmetric** (B_{sym}), **PSD**

($B_{\text{self_nonneg}}$), its **kernel** $\supseteq$ **constants** ($\text{lap_kills_constants}$ — row-sum-zero = conservation), **linear** (lap_linear), equals **div-grad** (lap_is_div_grad), and satisfies **summation-by-parts** (sbp).

- **Th 4.4 (energy non-increase).** the per-node equation of motion makes the readout energy

monotone non-increasing ($\text{list_energy_nonincreasing}$).

4.4 The constants of geometry are readout-invariants of quotient grammars — geometry closed

The recurring constants of geometry are **not** signatures imposed on nature, nor a catalogue of coincidences; each is the **invariant of a recurring quotient grammar** — it appears only after a structural reduction, and is refused where the reduction fails. (*Lahtee 2026, π, φ standalone synthesis; tiers below are exactly its claim-gate ledger.*)

- π = **the calibrated invariant of compact retained closure** (order-reversal witness Th_coqc -elig; the metric-calibration route is $+\mathbb{R}\text{-Open}$). Two roots,

one number: (i) an involutive nontrivial adjacent **order reversal** carries the exchange factor $r=-1=e^{i\pi}$ (Th 4.4, exact under the ordered-tape axioms; cyclic start-independence forces k **odd** — witness `formal/IDM_Harvest.v: odd_from_cyclic_closure/least_nontrivial_odd_is_three` [Toledo: L_R/M.15.v1] — the same k -odd witness that forces $k=3 \rightarrow \text{SU}(3)$, §II); (ii) a continuous compact transport closes on a **topological circle** $\theta \in \mathbb{R}/\mathbb{T}\mathbb{Z}$ and an **isotropic Euclidean metric calibration** gives $\mathbb{C}/\mathbb{R}=2\pi$. **This second route uses the completed real circle, so it is $+\mathbb{R}\text{-Open}$, not Th_coqc** — it is stated and predicted, not proved axiom-free (§10.13). The native, contamination-free content of π is route (i), the finite order-reversal phase; π is *not* inserted at the root — route (i) is what compact closure reads out, and route (ii) is its continuum readout.

- φ = **the Perron invariant of minimal closure-continuation transfer** (Th_coqc). The least

nontrivial two-step transfer $K_F=[[1,1],[1,0]]$ (Fibonacci) has characteristic $\lambda^2-\lambda-1=0$, so its Perron–Frobenius eigenvalue is $\varphi=(1+\sqrt{5})/2$ (Th 4.5, algebraically exact). Independently, the retained-history fusion sector (gates $H1\text{--}H7 \rightarrow$ a unitary fusion category) forces the minimal noninvertible frontier $\tau \otimes \tau \cong 1 \oplus \tau$ with $\text{FPdim}(\tau)=\varphi$ — a *fusion/history* dimension, **not** an ordinary carrier dimension (Cor 4.1: no finite-dimensional fibre functor, since $d^2=1+d$ has no integer solution). φ is irrational *because* it is a history dimension, not a count.

- **The golden rotation** $\theta_G=2\pi/\varphi^2 \approx 137.5^\circ$ (Th_coqc -elig: exact algebra + number theory). Combining a compact

phase (π) with the minimal-memory transfer (ϕ) gives nonresonant generation on a closed phase; its continued fraction $[1; 1, 1, \dots]$ is extremal for rational approximation (Hurwitz–Markov), which is the precise, Dr-fenced meaning of "most irrational" — it *delays* low-order phase locking. **Do not** read this as "every spiral/growth system is golden": the *universal golden nature* claim is **rejected** (negative controls: compact closure without minimal transfer gives π without ϕ ; minimal transfer on a line gives ϕ without π ; an arbitrary logarithmic spiral has a free growth rate; a Fibonacci count can be purely combinatorial).

- **Why the same constants recur across domains (the closure principle, Dr).** if domains D_a each

admit a translation $q_a: D_a \rightarrow Q$ to a shared structural quotient, and the observables factor through it ($O_a = \tilde{O} \circ q_a$), the *same* invariant appears in all of them **without any system transmitting a number to another** — recurrence is quotient-equivalence, not numerical transmission. " π is return; ϕ is growth" made precise: π = calibrated invariant of compact closure, ϕ = Perron invariant of minimal closure-continuation transfer.

Geometry closes here, uncontaminated (its *finite* content; the completed-manifold/topology frontier stays $+\mathbb{R}$ -Open, §10.13): distance is retained resistance (§4.1); betweenness is Tarski over the discrete metric (§4.2); angle is the overlap fraction, never a continuum degree (§4.2); curvature is the second retained difference with a discrete Gauss–Bonnet/Bianchi (§10.10); and its constants (π, ϕ, θ_G) are readout-invariants of quotient grammars — each derived, tier-fenced, and free of any injected continuum, Platonic constant, or "universal golden" overclaim.

Part V — The retained-information operator (information is the central axis)

5.1 The keystone (Def 5.1 + Th 5.1) [Toledo: Keystone]

- **Def 5.1 (retained-information density).** $I(\Phi) :=$ the distinguishability of neighbouring

retained states of the field Φ on the graph.

- **Th 5.1 (keystone — Th_coqc, witness formal/IDM_Keystone.v: keystone_B_eq_I).** $B(\Phi, \Phi) = I(\Phi) :$

the operator's Dirichlet energy is the retained-information functional. In the edge assembly of $L_R = D_W - W$ this is the identity $\Phi^T L_R \Phi = \sum_{\text{edges}} w \cdot (\Phi_i - \Phi_j)^2$, proved axiom-free over $\mathbb{Q}$ (Print Assumptions = *Closed under the global context*); its companion `keystone_nonneg` [Toledo: Keystone/M.07.v1] shows L_R is positive semidefinite (the retained metric is a genuine seminorm). *Information — not length, not energy — is the central quantity; geometry and distance are readouts OF it.*

5.2 Readouts of the operator

- **Th 5.2 (metric as readout).** geometry (distances, spectra) is recovered from L_R — geometry is

a readout of the information operator, not a prior stage.

- **Th 5.3 (quantum readout, $+\mathbb{R}$ carrier).** completely-positive trace-preserving (CPTP) channels

are the information-preserving quantum readout layer over the same field.

- **Pr 5.1 (mass/spectrum principle).** physical spectra and mass ratios are read from L_R 's

spectrum and from Born-rule overlap ratios — **never** from continuum angles, coordinates, or a smooth free-parameter coordinate (which carries no content: a smooth bijection of a knob is a coordinate, not an observable).

Part VI — The continuum as a readout, and the diagnosis of "open" problems

6.1 The diagnosis move (Pr 6.1, Dr)

When a problem involves infinity, **diagnose the injection first** (I1–I4 / Z1–Z4); the openness is usually its artifact. Then **dissolve** it (the continuum is not what appears) and **predict the discrete appearance** — a falsifiable statement — rather than defer to a non-readout.

- **Worked pattern (Navier–Stokes, Th_coqc bricks).** InfoGlobalLaminarReadout +

InfoDiffusionMaxPrinciple + InfoMonotoneMaxPrinciple $\Rightarrow$ global-in-time laminarisation; InfoSkewNotMonotone $\Rightarrow$ a skew (energy-conserving) advection is *never* monotone — so L^2 conservation and the L^∞ maximum principle cannot both come from one advection. That trade-off is the hardness, now a theorem; the continuum "blow-up" needs I2+I3+I4 even to be posed.

6.2 The endpoint principle (Th 6.1, Th_coqc for $\times$, Dr general —

InfoOperatorLosesPropertyAtEndpoints)

Every finite-property-preserving operator **loses a defining property at each refused endpoint**: at 0 it loses cancellation and invertibility; at ∞ there is no element to act on ($\mathbb{Q}$ has no maximal value). Consistent infinities in classical mathematics each *pay this price* (limits never touch ∞ ; $\aleph_0+1=\aleph_0$ drops cancellation; extended reals drop totality; ZFC drops unrestricted comprehension). So the discrete/finite interior is the **only** region where operators keep all their properties — by necessity, not taste.

Part VII — Complete operator reference (self-contained; no other math text is needed)

Every operator used anywhere is **defined here from the primitive**, with its domain, its recursion, its governing theorems, and its discrete-honest boundary (which inputs are refused non-readouts). Nothing is imported from an outside textbook — this Part closes the operator set.

7.0 The information language of every operator (the PRIMARY meaning — the old language is refused)

Principle (Pr 7.0, Dr). Every mathematical operator is, first, a **mode of retained distinction** — an operation on *readouts*, not on continuum quantities. Its ordinary textbook meaning ("add magnitudes", "multiply lengths", "the angle between two vectors") is only the machine **shadow**. This book gives each operator its **information meaning** as the primary reading; the recursions in 7.1–7.5 are merely its machine realization. **No operator keeps its old-world semantics.**

operator	information meaning (the new language — read this first)
$\neg$	obstruction reversal — flip which side of a distinction is retained
$\wedge$	joint retention — both distinctions held at once (the meet of two records)
$\vee$	admissible retention — either distinction suffices (the join)
$\rightarrow$	retained transport — a distinction at p carried, by a reader-transform, to one at q
$\leftrightarrow$	mutual transport — each record inferable from the other
$\forall$	retention across the whole tape — holds at every retained position
$\exists$	a constructible witness distinction — a reader can exhibit one
$=$	mutual indistinguishability — no reader can tell them apart ($0(x)=0(y)$ for every 0)
$\neq / \#$	a witnessed distinction — some reader does tell them apart
$<$	retained precedence — the arrow of accumulated record ("came before")
$\oplus (+)$	accumulation — merge two retained records into one longer record (total distinctions retained)

operator	information meaning (the new language — read this first)
$\ominus (-)$	cancellation / debt of retained distinction — remove record; below the ground the record becomes signed (a directed debt) — this is <i>what $\mathbb{Z}$ is</i>
$\otimes (\times)$	replication / composition of distinction-structure — lay b copies of a 's pattern end to end; the tensor of two retained patterns (distinctions of distinctions)
$\div$	equal partition of a retained record into b indistinguishable parts (the inverse of replication); $\div 0$ = partition into <i>no</i> parts = no reader-record = refused
$\wedge$	self-composition — iterate the replication-operator on its own output; the retained record of a process re-reading itself
$\sqrt{} / \sqrt[n]{}$	generator readout — recover the pattern whose self-composition yields the record; usually a non-readout (no finite record generates it), only its approach appears
$\log / \log_a$	retained-distinction count — how many replications (self-compositions) of a 's pattern produce the record; the record LENGTH; the native unit of information (bits when $a=2$) — inverse of $\wedge$
$\backslash$	$\cdot \backslash$ magnitude of retained distinction, direction discarded
Δ	local change of retained distinction (the increment of record between adjacent ticks)
Σ	accumulated retained record (the total distinction up to a tick)
$\langle \cdot, \cdot \rangle_G$	shared retained distinction — the co-retention of two records
overlap fraction	readable co-retention (Born-rule) — the only "angle", and it is not an angle

In one line: arithmetic is the **bookkeeping of accumulated, replicated, partitioned, and cancelled retained distinction**; logic is the **algebra of distinction combinators**; comparison is **reader-(in)distinguishability**. The continuum readings — length, magnitude, area, angle — are refused as primary and appear only as later readouts (Part III, VIII). (*This is the operator layer of the project's readout lexicon; keep the two in sync.*)

7.1 Logic operators (on RDL judgements — Part I)

- **$\neg$ negation** — $\neg p$ flips **p** 's obstruction polarity; NOT **ex falso** (Ax-RDL3). $\neg\neg p \vdash p$

only in the classical limit (Th 1.5).

- **$\wedge$ conjunction** — $p \wedge q$ retains both distinctions; $\text{obstruction}(p \wedge q) =$ the join of the two.

Commutative, associative, idempotent (Th_coqc , RDL.v).

- **$\vee$ disjunction** — $p \vee q$ retains either distinction; De Morgan up to obstruction.
- **$\rightarrow$ implication** — $p \rightarrow q :=$ the retained transformation carrying p 's distinction to q 's.

($p \rightarrow q \equiv \neg p \vee q$ **only** in the classical limit — do not assume it paraconsistently.)

- **$\leftrightarrow$ biconditional** — $(p \rightarrow q) \wedge (q \rightarrow p)$.
- **$\forall$ universal** (over D) — $\forall n. \varphi(n) := \varphi(0)$ and φ closed under succ (grounded in RD5

induction).

- **$\exists$ existential** — $\exists n. \varphi(n) :=$ a witness n is **constructible** (constructive existence; no

pure excluded-middle existence).

7.2 Equality and order operators

- **$=$ equality** — Def (readout-equivalence): $x = y \Leftrightarrow 0(x) = 0(y)$ for every readout 0

(indistinguishable to every reader); on D , structural via succ -recursion. Reflexive, symmetric, transitive, and **substitutive** (Leibniz). (Th_coqc .)

- **$\neq$ / $\#$ apartness** — $\neg(x=y)$; constructively, a genuine distinction is exhibited (on $\mathbb{R}$, apartness $x \# y$ is primitive, $\neq$ its negation-form).

- **$<$ / $\leq$ order** — Def on D : $a < b \Leftrightarrow \exists k \neq 0, a \oplus k = b$. Total, well-ordered (Th 2.3). Lifted

compatibly to $\mathbb{Z}$, $\mathbb{Q}$; on $\mathbb{R}$ the strict Rlt is **cotransitive** (Th 3.7) and Rle a partial order — **no total $\leq$ constructively** (Pr 3.1).

- **$< \leq > \geq$** := the $< / \leq$ instances per number system.

7.3 Arithmetic operators (each defined from succ , recursively)

- **$\oplus$ addition ($+$)** — $a \oplus 0 := a$; $a \oplus \text{succ}(b) := \text{succ}(a \oplus b)$. **Th 7.1:** commutative, associative, cancellative, 0 identity (semiring, Th 2.2). Lifted to $\mathbb{Z}/\mathbb{Q}/\mathbb{R}$ as $+$.

- **$\ominus$ subtraction ($-$)** — total on $\mathbb{Z}$ as $a \ominus (-b)$, $-b$ the additive inverse (Grothendieck,

Th 3.1). On $\mathbb{D}$ it is **PARTIAL** (truncated: $a \ominus b$ defined only when $b \leq a$) — *subtraction below zero is refused; that refusal is exactly where $\mathbb{Z}$ is born.*

- **$\otimes$ multiplication ($\times$)** — $a \otimes 0 := 0$; $a \otimes \text{succ}(b) := (a \otimes b) \oplus a$. **Th 7.2:** commutative, associative, distributes over $\oplus$, **1** identity (semiring). Lifted to $\mathbb{Z}$ (ring), $\mathbb{Q}/\mathbb{R}$ (field).
- **$\div$ division** — on $\mathbb{Q}$: $a \div b := a \otimes b^{-1}$, b^{-1} the multiplicative inverse (Th 3.4, `qmul_inv`),

defined iff $b \neq 0$. $\div 0$ is REFUSED — the endpoint principle (Th 6.1): at 0 the operator loses invertibility; $1/0$ is the reciprocal non-readout to ∞ (Pr 0.2), never a value. On $\mathbb{D}/\mathbb{Z}$: **Euclidean division** $a = q \otimes b \oplus r$ with $0 \leq r < b$ (quotient + remainder), total for $b \neq 0$.

- **$\wedge$ exponentiation (ยกกำลัง)** — $a \wedge 0 := 1$; $a \wedge \text{succ}(n) := a \wedge n \otimes a$ (iterated $\otimes$). **Th 7.3:**

$a \wedge (m \oplus n) = a \wedge m \otimes a \wedge n$, $(a \wedge m) \wedge n = a \wedge (m \oplus n)$, $(a \otimes b) \wedge n = a \wedge n \otimes b \wedge n$. Integer exponents on $\mathbb{Q}$: $a \wedge (-n) := 1/a \wedge n$ ($a \neq 0$). **Rational / real exponents are a READOUT:** $a \wedge (p/q)$ is the root (7.3 root); $a \wedge x$ for $x \in \mathbb{R}$ is the limit of $a \wedge (\mathbb{Q}$ -approximants) — the finite approximants appear, the "value" is a readout-invariant. Convention $0 \wedge 0 := 1$ (empty product), stated.

- **$\sqrt[n]{}$ / $\sqrt[n]{}$ root (ราก)** — $\sqrt[n]{a} :=$ the r with $r \wedge n = a$ ($r \geq 0$ for even n). **Th 7.4:** when a is an n -th power in $\mathbb{Q}$, $\sqrt[n]{a} \in \mathbb{Q}$ (exact readout). **Otherwise $\sqrt[n]{a}$ is a NON-READOUT** (Th A.1, `InfoIrrationalNonReadout`: $\sqrt{2} \notin \mathbb{Q}$) — only its regular Cauchy sequence of $\mathbb{Q}$ -approximants appears; " $\sqrt{2}$ the number" is a boundary, never an appearance. It is *computed* by a discrete bisection/Newton sequence over $\mathbb{Q}$ that converges by Cauchy-completeness (Th 3.6) — no $\mathbb{R}$ is needed to run it.

- **$\log_a$ logarithm — the retained-distinction count (the information unit)** — Def: $\log_a(x) :=$

the n with $a \wedge n = x$, i.e. **how many self-compositions of a generate the record x** — the inverse of $\wedge$. On $\mathbb{D}/\mathbb{Z}$: the **discrete logarithm** $\lfloor \log_a x \rfloor =$ the largest n with $a \wedge n \leq x$, total for $a \geq 2$, $x \geq 1$. **Th 7.5:** $\log_a(x \otimes y) = \log_a x \oplus \log_a y$, $\log_a(x \wedge k) = k \cdot \log_a x$, $\log_a(1) = 0$ — replication becomes accumulation ($\otimes \rightarrow \oplus$): the log **linearises composition**. When x is not an exact power of a , $\log_a x$ is a **non-readout** (only its $\mathbb{Q}$ -approximant appears, computed by a discrete bisection over exponents). **Information meaning (primary):** $\log_2 N$ is the number of **binary distinctions (bits)** needed to name one of N retained states — the record LENGTH; the entropy $H = \sum p_i \cdot \log_2(1/p_i)$ (Part V) is the *expected retained-distinction count*. $\log$ is therefore the native unit of this book's "information" — it measures retained distinction in bits. $\log_a(0)$ and $\log_a$ of a non-positive record are **refused** (no generating count exists — the $0/\infty$ endpoint, Th 6.1).

7.4 Derived operators (all reduce to 7.1–7.3)

- $|\cdot|$ **absolute value** — $|a| := a$ if $a \geq 0$ else $\ominus a$; non-negative, $|a \oplus b| \leq |a| \oplus |b|$.
- **gcd / lcm** — from Euclidean division (7.3 $\div$).
- Δ **difference**, Σ **sum** — the discrete-calculus operators (Th 3.2–3.3), inverse to each other.
- $\langle \cdot, \cdot \rangle_G$ **inner product**, $\|\cdot\|_G$ **norm** — $\|x\|_G := \sqrt{\langle x, x \rangle_G}$ (root over $\mathbb{Q}$, a readout if

irrational); the **overlap fraction** (Def 4.2a) is the ONLY angle-like operator — trig-free.

7.5 The refused operations (the endpoint boundary, made explicit — Th 6.1)

$\div 0$ (loses invertibility) · a supremum/exponent reaching ∞ ($\mathbb{D}$ has no maximal element)
 · $\sqrt[n]{}$ of a non- n -th-power **as an exact value** (a non-readout) · any limit that "lands" at ∞ or an exact 0 . All refused as non-readouts; only the finite $\mathbb{Q}$ -approach appears. **This closes the operator set:** every total operation is defined from **succ**; every partial one has its refused inputs named — so no external mathematics is required to evaluate any expression in this book.

Part VIII — Information Discrete Calculus (the calculus of retained difference; [results-first])

The calculus built entirely on the difference Δ and sum Σ operators over $\mathbb{Q}$ (no reals) — the discrete analogue **and root** of continuum calculus; continuum calculus is recovered last, as a readout. Results and definitions are stated here; proofs are v-proofs (next version).

8.1 Sequences, shift, difference

- **Def.** a sequence is a map $f : D \rightarrow F$ ($F = \mathbb{Q}$ or a field). Shift $E f(n) := f(n+1)$.
- **Def.** forward difference $\Delta f := (E-1)f$, i.e. $\Delta f(n) = f(n+1) \ominus f(n)$; backward $\nabla f(n) = f(n) \ominus f(n-1)$;

higher Δ^k . **Newton forward-difference (discrete Taylor):** $f(n) = \sum_{k=0}^n \binom{n}{k} \Delta^k f(0)$ — exact, finite, no limit.

8.2 Sum, the discrete FTC, and integration-by-parts

(Machine-checked core: the exact D_{ε} sum/scalar/product rules, FTCC telescoping, and summation-by-parts are `formal/IDM_Calculus.v`, axiom-free — see §10.5 Th 10.8.)

- **Def.** antidifference $\Sigma_a^b f := f(a) \oplus \dots \oplus f(b-1)$.
- **Th 8.1 (discrete FTC).** $\Sigma_a^b \Delta f = f(b) \ominus f(a)$ (telescoping) and $\Delta(\Sigma f) = f$ (Th 3.2).
- **Th 8.2 (summation by parts).** $\Sigma f \cdot \Delta g = f \cdot g \ominus \Sigma (Eg) \cdot \Delta f$ — discrete integration by parts.

8.3 Falling powers — the natural discrete monomials

- **Def.** falling factorial $n^{\{(k)\}} := n(n-1)\dots(n-k+1)$, $n^{\{(0)\}} := 1$.
- **Th 8.3 (discrete power rule).** $\Delta n^{\{(k)\}} = k \cdot n^{\{(k-1)\}}$ and $\Sigma n^{\{(k)\}} = n^{\{(k+1)\}} / (k+1)$ —

the clean power rule holds for **falling** powers (WHY they, not ordinary powers, are the discrete monomials). Stirling numbers convert ordinary $\leftrightarrow$ falling powers.

8.4 Difference equations (the discrete ODEs)

- **Def.** linear difference equation $\Sigma_i c_i f(n+i) = g(n)$; solved by characteristic roots

(homogeneous) + a particular part.

- **Def.** the discrete exponential 2^n (eigenfunction of E) and the retained growth law

$\Delta f = a \cdot f \Rightarrow f(n) = f(0)(1+a)^n$ — the discrete root of e^{ax} ; the continuum e is its $h \rightarrow 0$ readout ($+\mathbb{R}$).

8.5 Formal series, generating functions, discrete transforms

- **Def.** ordinary / exponential **generating functions** as $\mathbb{Q}$ -formal power series (formal — no

convergence, hence no continuum needed).

- **Def.** the **Z-transform** (discrete Laplace); the **discrete Fourier** on n -th roots-of-unity read out as **overlap fractions / turning numbers** (Def 4.2) — never continuum angles.

8.6 The calculus of the graph operator L_R (discrete exterior calculus)

- **Def.** coboundary d , its adjoint δ , gradient/divergence/curl, the **Hodge** decomposition and

harmonic forms — the discrete de Rham complex over $\mathbb{Q}$.

- **Def.** discrete **Green's function** of L_R ; the **heat step** $(1 - \tau L_R)$; the **spine wave** $M\partial^2_t + D\partial_t + K \cdot L_R$ (Part V's operator in motion) — all exact over $\mathbb{Q}$.

8.7 Continuum calculus as the last readout ($+\mathbb{R}$ -axioms)

- The continuum derivative $f'(x) = \lim_{h \rightarrow 0} \Delta f / h$ and integral $\int$ are recovered as **readouts** of

Δ / Σ under $h \rightarrow 0$ sampling — the final rung, flagged $+\mathbb{R}$, never the primitive. `InfoContinuumLimit`.

8.8 The discrete Jacobian (retained sensitivity), readout collision, and the retention lift

For a readout map $F: \mathbb{Q}^m \rightarrow \mathbb{Q}^k$ the **discrete Jacobian** J_F is the exact rational matrix of retained sensitivities $J_F[i, l] = \partial F_i / \partial x_l$ — the multivariate D_ϵ , computed by exact-algebra with no limit. It is the *retained-distinction sensitivity* of each readout coordinate to each source coordinate.

- **Local distinction-preservation.** $\det J_F \neq 0$ means F is locally injective (étale) — no retained

distinction is lost to first order. This can hold *everywhere and be exact*: the polynomial map $P = (1+xy)^3 z + y^2(1+xy)(4+3xy)$, $Q = y + 3x(1+xy)^2 z + 3xy^2(4+3xy)$, $R = 2x - 3x^2 y - x^3 z$ has the **constant** $\det J_F = -2$ (`Th_coqc-elig` / exact-algebra; validated `validation/discrete_jacobian.py`). A constant nonzero Jacobian determinant is a purely finite, conjecture-free readout.

- **Readout collision.** Local injectivity does **not** give global injectivity: the same F sends the

three distinct sources $(0, 0, -\frac{1}{4})$, $(1, -3/2, 13/2)$, $(-1, 3/2, 13/2)$ all to the **one** readout $(-\frac{1}{4}, 0, 0)$. A readout can collapse distinct sources — it *loses distinction* globally while preserving it locally.

- **The retention lift.** The distinction the readout discarded is restored by ****retaining** an extra

coordinate** ψ : the lifted map $s \mapsto (F(s), \psi(s))$ is injective exactly where F alone collides ($\psi \in \{-1, 0, 1\}$ separates the three). *Retention is the injectivity a readout loses* — the discrete, exact-algebra form of "carry the retained coordinate" (D_r ; the **det**, collision, and lift are all finite exact facts). (*Bridge to standard mathematics: this is a polynomial endomorphism with constant Jacobian — the setting of the Jacobian-conjecture literature — read here through retained distinction: the map is étale, the fibres are finite, and the retained lift trivialises the fibre.*)

8.9 The information reading (why it is *information* calculus)

Δ = local change of distinction, Σ = accumulated retained record, L_R = information diffusion; energy is a readout of the information functional (Th 5.1, $B(\Phi, \Phi) = I(\Phi)$). **Discrete calculus is the calculus of retained-information flow** — continuum calculus is what a finite reader reconstructs from it.

The unifying stepper (a discrete damped wave on the graph). The results of §5, §8, and §21.3 are one operator: $\mathcal{M} := M \cdot \partial^2 \epsilon + D \cdot \partial \epsilon + L_R$ on a finite weighted graph ($M, D > 0$ scalars) — a **discrete telegrapher's / damped-wave equation** $M \Phi'' + D \Phi' + L_R \Phi = S$. Its structure is forced from the root: normalising by M gives $\Phi'' + (D/M)\Phi' + (1/M)L_R \Phi = S/M$: the second-order term makes it **hyperbolic** (finite propagation — per L_R -mode λ the dispersion scale is $\sqrt{(\lambda/M)}$, not a single scalar speed), the **memory/relaxation time is the forced ratio** $\tau_c = M/D$ (a readout, not a free dial — set by the damping, not the speed), the first-order term gives strict energy decay, and $L_R \geq 0$ (the keystone, §5.1 **keystone_nonneg** [Toledo: Keystone/M.07.v1]) makes the homogeneous energy non-increasing (**relaxation_dissipation** [Toledo: Keystone/M.08.v1]). So one graph operator carries diffusion, waves, damping, and the no-blow-up bound — a single discrete object, no continuum. (*Extracted from the readout-genesis master-equation forcing ledger; told here as the pure mathematics of the operator $M\partial^2 + D\partial + L_R$, tiers as in §5/§21.3.*)

Part IX — Powerful Tools, Re-adapted (Regularization, Asymptotics, Transforms, Deformations)

Under readout-not-truth there is no completed infinite object to sum, integrate, or continue; what exists is a finite reader's *retained aggregation* of discrete data at a declared resolution. Every tool below is therefore re-read as one operation — split a finite readout into a **window-tracking bulk** (discarded, non-readout) and a **window-stable residue** (kept), and declare the limit only after proving stability (Discipline **A8**). Regularization is not "assigning a value to infinity"; it is the mathematics of finite retained aggregation.

A. Divergent aggregation as retained readout

§9.1 Ramanujan summation. The Ramanujan value is not the sum of the series but the resolution-independent residue of the finite prefix-sum $I1f[N] = \sum_{n \leq N} f[n]$. Factor $I1f[N] = A \operatorname{varepsilon}(N) \oplus C \operatorname{varepsilon}(f) \oplus R_p(N)$, with $A \operatorname{varepsilon}$ the *window-growing envelope (integral + boundary derivative corrections)*, $R_p(N) \rightarrow 0$ the truncation error, and $C \operatorname{varepsilon}(f)$ the *stable residue*. Declare $R \neq \sum f = C \operatorname{varepsilon}(f)$ only after proving $cN = I1f[N] - A \operatorname{varepsilon}(N)$ is *Cauchy under window-refinement*. Buys the flagship $R \neq \sum n = \zeta(-1) = -1/12$ as a finite computation, with the firewall against " $1+2+3+\dots = -1/12$." **Tier: finitediagnostic.**

§9.2 Euler–Maclaurin. The exact (Rm -attached) sum $\leftrightarrow$ integral bridge, re-founded as the first nontrivial corollary of FTCC ($\operatorname{I} \operatorname{varepsilon} D \operatorname{varepsilon} f = f$) via the finite operator algebra $E \operatorname{varepsilon} = (1 - \operatorname{varepsilon} D \operatorname{varepsilon})^{-1}$: $E \operatorname{varepsilon} f[a, b] = \frac{\operatorname{varepsilon}^2}{2} [f(b) + f(a)] + \sum_{k=1}^m \frac{B_{2k}}{(2k)!} \operatorname{varepsilon}^{2k} \left[D \operatorname{varepsilon}^{2k-1} f(b) - D \operatorname{varepsilon}^{2k-1} f(a) \right] + R\{m, \operatorname{varepsilon}\}$. *Bernoulli numbers are finite rational data; the divergent $m \rightarrow \infty$ tail is un-coarse-graining without bound, so truncating at retained order $m^{\ast}$ is a $G\{\lambda(m^{\ast})\}$ choice.* This is the constructive engine realizing the regularization hook at every finite order. **Tier: finite_diagnostic** (interpretive layer Dr).

§9.3 Zeta / analytic regularization. Build the two-parameter readout $\zeta_{\lambda}(s; N) = \sum_{n \leq N} n^{-s} e^{-\lambda n}$ — manifestly $E = \tilde{E} \circ G_{\lambda}$ with $G_{\lambda} = e^{-\lambda}$ discarding the sub-resolution tail (S1). Euler–Maclaurin splits it into λ -divergent poles plus a regulator-independent residue; when regulator-independence is *proved*, that residue is $\zeta(-s)$. The operator version regularizes $\det A = \prod \lambda_i$ as $\det \zeta A = \exp(-\zeta A'(0))$. *The pole at $s=1$ is the framework correctly refusing a readout where none is stable.* **Tier: finitediagnostic** (physical application Dr).

§9.4 Abel, Borel, Cesàro. Each regulator is a coarse-graining parameter λ . Cesàro: $CN = \frac{1}{N} \operatorname{varepsilon}(s)[N]$, a finite mean; report its plateau. Abel: $Gx[a][n] = a[n] x^n$ suppresses indices past $n^{\ast} \sim 1/\ln(1/x)$. Borel: $BN(t) = \sum_{n \leq$

$N\} \frac{a_n}{n!} t^n$ then discrete aggregation $S|varepsilon = |varepsilon| \sum_k e^{-k|varepsilon|} B(k|varepsilon)$. All read: the family $\{E[\lambda[a]]\}$ has a stable value SS once $|\lambda|$ crosses a declared threshold; the raw sum is never computed. The canonical teaching example for why A8 exists (forget the stability declaration and "prove" $1-1+1-\dots$ is both 0 and 1). **Tier: finitediagnostic.**

§9.5 Generating functions. $\mathbb{Q}[[x]]$ is already finite-discrete: it is the inverse limit of decidable rings $\mathbb{Q}[x]/(x^{N+1})$, with GN truncation and $x^n = \sum_{k \leq n} a_k b_{n-k}$ a finite sum. Multiply-by- x is the causal shift ($I|varepsilon$ -side), $(A-a_0)/x$ the backward-difference dual; solving a linear recurrence via $A(x) \cdot (1 - \sum c_k x^k) = \text{data}$ is FTCC made combinatorial. Evaluating a rational closed form at a divergence point x_0 is the same regularization move — licensed only post-stability. **Tier: finite_diagnostic** (evaluation-at-a-point Dr).

§9.6 Ramanujan Master Theorem, mock theta, partition asymptotics. RMT is A8 on the coefficient axis: from retained $\{\phi(0), \dots, \phi(N)\}$ form $f|varepsilon(x)$, take the finite Mellin readout $M|varepsilon(s)$, and declare $\Gamma(s)\phi(-s)$ only after a growth-bound stability proof. Hardy–Ramanujan–Rademacher $p(n) \sim \frac{1}{4n\sqrt{3}} e^{\pi\sqrt{2n/3}}$ is the case A8 passes — Rademacher's convergent tail certifies the discard. Mock theta functions are the case it fails: the sub-leading discrepancy from modularity survives every finite resolution and must be carried explicitly as Zwegers' shadow. Gives A8 falsifiable teeth. **Tier: finite_diagnostic.**

B. Asymptotics and transforms

§9.7 Mellin transform & master theorem. Read q —no, read s as a probe-exponent and the transform as a coarse-graining on a geometric log-grid: $M|varepsilon[f] = \sum_n f|varepsilon[n] x^{n-1} (x^n - x^{n-1})$, an $I|varepsilon$ aggregation. The "fundamental strip" becomes the **retained-distinguishability band** (found by A8 stability testing). The harmonic-sum factorization $M|varepsilon[S] = F^{\ast}(s) M|varepsilon[f]$ with $F^{\ast}(s) = \sum_k \lambda_k \mu_k^{-s}$; contour-shift = resolution change, residues = zeta/Ramanujan-type finite retained values. **Tier: finite_diagnostic.**

§9.8 Watson, Laplace, stationary-phase, saddle-point. Set $|varepsilon| = 1/\lambda$; the large parameter is inverse resolution. Discretize: $I|varepsilon = |varepsilon| \sum_{n \leq N} f(n|varepsilon) e^{-n} = \sum_k a_k |varepsilon|^{k+1} SN(k)$, $SN(k) = \sum_n n^k e^{-n}$ with a provable geometric tail (S1 discard). Declaring the bound then $N \rightarrow \infty$ recovers Watson's $\sum a_k k! / \lambda^{k+1}$ — each term labeled a coarse-graining moment. Laplace/stationary-phase use the finite argmin of $D|varepsilon^2 \phi$ and a discrete Gauss sum; the divergent asymptotic series ties back to §9.1–9.4. **Tier: finite_diagnostic** (saddle-point in $\mathbb{C}$, general optimal-truncation: Dr/Open).

C. Deformations and finite representations

§9.9 Umbral & finite-difference calculus. On the $|varepsilon$ -falling-factorial basis $x^{\underline{n}}|varepsilon = \prod_{j < n} (x-j|varepsilon)$ the causal difference is exact, not

limiting: $D \backslash \varepsilon x^{\{\underline{n}\}} \backslash \varepsilon = n \backslash, x^{\{\underline{n-1}\}} \backslash \varepsilon$, giving exact telescoped power-sums and the Newton forward reconstruction $f(x) = \sum n \frac{x^{\{\underline{n}\}} \backslash \varepsilon}{n! \backslash \varepsilon^n} D \backslash \varepsilon^{\{(n)\}} f[x_0]$. Stirling numbers are the resolution-deformation dictionary between x^n and the readout-native basis; B_n measures the basis-correction Faulhaber needs. **Tier: mixed** — finite core Thcoqc-eligible; the $\zeta(-n) = -B_{n+1}/(n+1)$ bridge is an import (Dr/Open), never merged with it.

§9.10 q-calculus. Same causal scaffolding on a geometric lattice $\Lambda_q = \{x_0 q^n\}$ — resolution measured as a ratio, not an absolute step. $Dq[n]$ is $D \backslash \varepsilon$ with local $\varepsilon n = x_0 q^{n-1}(1-q)$; $Iq f(a) = a(1-q) \sum q^n f(aq^n)$ is $I \backslash \varepsilon$ with the q^n factor making the "infinite" sum a genuinely finite retained aggregate (S1 tail explicit). $q \rightarrow 1$ is the disciplined continuum recovery, proved not assumed. **Tier: finite_diagnostic.**

§9.11 Continued fractions. The CF algorithm is the access operator: each step emits one integer $a_k = \lfloor x_k \rfloor$, $x_{k+1} = 1/(x_k - a_k)$, aggregated by the prefix matrix product $M_n = \prod \begin{smallmatrix} a_k & 1 \\ 1 & 0 \end{smallmatrix}$. Depth n is λ ; p_n/q_n is the optimal finite rational readout at denominator budget q_n , with $|x - p_n/q_n| < 1/(q_n q_{n+1})$ as the resolution-deformation metric. Periodicity (Lagrange) is itself a finite stability certificate. **Tier: finitediagnostic** (finite n Thcoqc; general $n \rightarrow \infty$ + $\mathbb{R}$ (Open)).

§9.12 Padé approximants. A rational-function sibling of the scalar regularizers: from a retained coefficient window $(c_0 \backslash \varepsilon, \dots, c_N \backslash \varepsilon)$, solve the finite Padé linear system for $[L/M]$. Poles geometrize S1's discarded tail as singularity locations. Convergence as $N \rightarrow \infty$ holds only under structural hypotheses (de Montessus); the diagonal is genuinely **Open** for unclassified f (Baker–Gammel–Wills refuted, Lubinsky 2003). **Tier: split** — construction Th_coqc, convergence Dr/+ $\mathbb{R}$ (Open); spurious Froissart poles are artifacts, never reported as structure.

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Every tool above is re-adapted to this framework (finite- ε , coarse-graining, A8) and the original is cited; none is used in its continuum-primitive form. This is the operator/analysis power layer — the book carries the world's hard tools without importing the continuum.

Part X — Integrated extensions (sets & functions, cardinality, analysis, probability, reconciliations)

This Part closes the foundation's remaining gaps and reconciles it with the external record. **Honest global status:** the tractable finite/decidable results in §10.1–10.4 now carry **actual formal/*.v witnesses** (Th 10.1 `kuratowski_pair_inj` [Toledo: `delta_R/M.02.v1`]; Th 10.2 `formal/IDM_SetsFunctions.v` — **the function $\cong$ functional-relation coincidence, converted from a sketch in the v-proofs edition**; Th 10.3–10.5 `IDM_FiniteWitnesses2.v`; Th 10.6 propositional `IDM_Logic.v` and **first-order formal/IDM_FirstOrder.v** — **the finite-domain quantifier recursion, also converted in the v-proofs edition**; Th 10.7 `IDM_Logic.v`), each confirmed **Closed** under the global context by `bash formal/verify.sh`. What remains a *declared* frontier is §10.5's $\varepsilon \rightarrow 0$ real-analysis quantifier and §10.6's completed-measure limit — genuinely **+ $\mathbb{R}$ -Open** (they quantify over a completed continuum), honestly fenced, not dressed as `Th_coqc`. No external validation is invoked (horizontal-knowledge policy).

10.1 The pair-product-relation-function layer, from δ_R (closes the biggest "not-standalone" blocker)

Built strictly from prior primitives (number tower, G_λ -fibers, $\sim_\lambda$, δ_R) — a strict DAG, no forward reference. **Base already machine-checked (economize):** the semiring the pair layer stands on is `solver-arc-private/formal/RD.v` (sister repo, not local — `RD3_succ_ne_zero`, `RD4_succ_inj`, `add_assoc`, `add_comm`, `mul_add` distributive, `Th_coqc`); only the Kuratowski pair / product / function step below is the new sketch.

- **Singleton / pair fibers.** for a, b in fiber X at resolution λ : $\{a\}_\lambda$, $\{a, b\}_\lambda$ are

sub-fibers selected by finitely many $\sim_\lambda$ -classes (fiber-formation is already licensed, §0.5.3).

- **Ordered pair (δ_R -Kuratowski).** $(a, b)_\lambda := \{\{a\}_\lambda, \{a, b\}_\lambda\}$; degeneracy $a =_\lambda b$ is *detected*

as a δ_R count ($|outer|=1$ vs 2), not hidden. **Th 10.1 (injectivity, `Th_coqc` — witness `formal/IDM_FiniteWitnesses.v: kuratowski_pair_inj`):** $(a, b) = (c, d) \iff a =_\lambda c \wedge b =_\lambda d$, by a finite case-split on $|\{\cdot, \cdot\}| \in \{1, 2\}$ — every step a decidable finite readout.

- **Cartesian product.** $X \times_\lambda Y := \{(a, b)_\lambda : a \in X, b \in Y\}$ by finite enumeration (A8: no infinite

product; $\lambda \rightarrow 0$ deferred **+ $\mathbb{R}$ -Open**); $|X \times Y| = |X| \cdot |Y|$.

- **Relation** = an admissible sub-fiber of the product — literally $E = \tilde{E} \circ G_\lambda$ applied to $X \times Y$.

- **Function** = a relation R with $\text{IsFunction}(R) := \forall a \in X \exists!_{\lambda} b \in Y, (a,b) \in R$ ($\exists!_{\lambda} a$ finite)

decidable search). **Th 10.2 (coincidence — closes the blocker, Th_coqc, witness formal/IDM_SetsFunctions.v):** $\text{AdmissibleMap}(X,Y) \cong \{R : \text{IsFunction}(R)\}$ via the graph $G(E)$. Machine-checked: the graph of a map on a finite domain is **total** (`graph_total` [Toledo: deltaR/M.07.v1]) and **single-valued** (`graph_single_valued` [Toledo: deltaR/M.08.v1]) — i.e. a function; the correspondence is **faithful**, maps agree on the domain iff their graphs coincide (`graph_faithful` [Toledo: deltaR/M.09.v1] / `graph_faithful_converse` [Toledo: deltaR/M.10.v1]); and the converse construction recovers a map from any functional relation (`functional_relation_has_map` [Toledo: delta_R/M.11.v1]). **Axiom-free** (confirmed Closed under the global context): all objects are finite lists over a decidable equality; only `List` + decidable equality is used — no LEM / choice / funext.

10.2 Cardinality and the potential-vs-actual infinite

(In-framework; type-level refusal is a framework-native observation.)

- **Count = retained distinctions.** $\text{Counting}_{\lambda}(S) := |[S]_{\lambda}|$ (de-duplicated $\sim_{\lambda}$ -classes) — always

a finite natural, never a primitive ∞ .

- **Same size** = an admissible bijection $\approx_{\lambda}$. **Th 10.3 (Th_coqc, witness formal/IDM_FiniteWitnesses2.v: same_set_same_size):** for `NoDup` lists,

same members $\Rightarrow$ equal count (equinumerosity = admissible bijection), by finite induction — axiom-free.

- **Potential infinite** = a δ_R -generated tape $\text{Tape}(n+1) = \sigma(\text{Tape } n)$, each stage finite with

$\text{Counting}(\text{Tape } n) = n+1$. **Th 10.4 (Th_coqc, witness formal/IDM_FiniteWitnesses2.v: tape_count_succ / tape_no_terminal** [Toledo: D/M.44.v1]): strict growth $\Rightarrow$ no terminal stage (an unbounded *process*, no completed object).

- **Actual infinite refused — analytically.** **Th 10.5 (Th_coqc, witness formal/IDM_FiniteWitnesses2.v: no_infinite_readout** — the witness proves the *finite-length* fact $\forall l:\text{list } A, \exists n, \text{length } l = n$; the exclusion of an *infinite inhabitant* is the structural property of the inductive type `list A`, not this lemma):

Every readout inhabits `list A`, which has *no infinite inhabitant* — actual ∞ is excluded because the readout type never had room for it, not by fiat. (Cantor-style comparison of two tapes' growth is per-schema `finite_diagnostic`, not one closed theorem.)

10.3 Finite satisfaction $\models_\lambda$ (and an honest downgrade)

- **Th 10.6** (**Th_coqc** — propositional witness `formal/IDM_Logic.v: finite_satisfaction_dec / sat_dec` [Toledo: D/M.57.v1], decidable model-checking over $\neg\forall$; **FIRST-ORDER** witness `formal/IDM_FirstOrder.v: eval_correct / sat_fo_decidable` [Toledo: D/M.53.v1] — the quantifier-over-finite-domain Tarski recursion $\forall\rightarrow\text{forallb}$, $\exists\rightarrow\text{existsb}$, proved sound+complete and decidable, axiom-free): $\models_\lambda$ — a decidable, computable Tarski recursion of satisfaction on a

finite domain + finite formula at a fixed coarse-graining level. **Existing witness to REUSE (economize):** the readout-bivalence consistency layer is already machine-checked — sister repo `solver-arc-private/formal/RD_ConPA_ReadoutBivalence.v` (`soundnessC_param`, `consistencyC_param`, `Con_PA_classical_param`, `Th_coqc` — verified directly, `Print Assumptions` reports *Closed under the global context* for both, no `classic`); only the general finite $\models_\lambda$ recursion is the new sketch, and it builds on that file.

- **Downgrade (honest, tier note — see also §2.3 Th 2.5/Th 2.6):** the earlier " $D \models PA$ " (full

first-order Peano) and " $D \equiv \mathbb{N}$ " (elementary equivalence / nonstandard models) are flagged **+R-Open** here — they quantify over the completed standard model, and are NOT *reproven from scratch in this local §10.3 development*; only $\models_\lambda$ is a new claim here. This is a philosophical downgrade about completed-totality quantification, not a claim that the underlying Coq proofs (`sat_transfer / sentence_transfer / D_models_PA` in `solver-arc-private/formal/RD.v`) are unverified — they are machine-checked axiom-free (§2.3). Whether a Coq-checked $\forall$ -statement over D still counts as smuggling a completed totality under this repo's own readout-first stance is an open tier call, stated here rather than resolved silently in either direction.

10.4 RDL boundary-tie — making the logic load-bearing

- **Th 10.7** (**RDL non-explosion is** `Th_coqc`, witness `formal/IDM_Logic.v: rdl_non_explosion` — a Belnap–Dunn 4-valued countermodel where $p \wedge \neg p$ is designated but q is not, with `classical_would_explode` [Toledo: `delta_R/M.05.v1`] as the 2-valued control; that D 's semiring closure *requires* this is the **Dr** boundary-tie): D 's semiring closure *requires* RDL's non-explosion, not classical

logic — a boundary distinction (a state at the semiring frontier) is held without collapse only because `ex falso` is refused (Ax-RDL3). This is the one rung that makes Part I (logic) *load-bearing* for Part II (number), answering the review's "RDL is decorative" finding.

10.5 Real analysis via A8 (the finite- ε rung skeleton)

Every base definition is finite- ε over $\mathbb{Q}$; the $\varepsilon \rightarrow 0$ continuum statement is gated behind a **named, declared** stability proof (finite- ε Cauchy for limits/series/derivatives; discrete Euler–Maclaurin for integral tails). The uniform pattern below, repeated five times, is the rung: (1) a *decidable* finite- ε predicate over $\mathbb{Q}$, (2) a *stability witness* (a computable modulus map, or an explicit Euler–Maclaurin remainder bound), (3) only then a *declared* $\varepsilon \rightarrow 0$ readout — never a bare limit primitive. This closes the panel-flagged stub with all five sub-rungs stated, not just sketched.

10.5.1 Sequence limit (finite- ε Cauchy first)

- **Def 10.5a (ε -stability).** For $a : D \rightarrow \mathbb{Q}$ and $\varepsilon \in \mathbb{Q}_{>0}$, a is **ε -stable from N** iff $\forall m, n \geq N, |a_m - a_n| \leq \varepsilon$. $N_\varepsilon(a) :=$ least such N , when it exists — a **decidable, bounded search** for any concrete computable bound, hence checkable per (a, ε) instance. **Tier: `finite_diagnostic`**.

- **Th 10.10 (declared-stability limit, `Th_coqc-elig scaffold / +R-Open` at the quantifier).** If

$\varepsilon \mapsto N_\varepsilon(a)$ is itself an admissible (computable) *modulus map* — witnessed, not merely asserted to exist for every ε — define the readout limit $L(a)$ as the $\sim_\lambda$ -class of the eventual band $\bigcap_{\varepsilon} [a_{N_\varepsilon} - \varepsilon, a_{N_\varepsilon} + \varepsilon]$. This is the $\varepsilon \rightarrow 0$ statement, licensed **only** once the modulus map is exhibited — the Bishop "Cauchy-with-modulus" move (§10.7), re-derived here as the A8 stability declaration. **Per-witnessed-instance: `finite_diagnostic`**. The unrestricted " **$\forall$ sequence $\exists$ modulus**" existence claim (classical Cauchy completeness with no witness supplied) stays **`+R-Open`** — ties to §10.13.

10.5.2 Series convergence (partial sums + Euler–Maclaurin tail)

- **Def.** $S_n := I_1 a[n]$ (§8.2 antidifference / discrete FTC) — the finite partial sum, already

primitive, no new object.

- **Th 10.11 (series = sequence-limit stability of (S_n)).** Convergence of $\sum a$ is by *definition*

ε -stability (10.5a) of the partial-sum sequence (S_n) — no independent series primitive is introduced, closing a common redundancy. **Tier: `finite_diagnostic`**, same status as 10.5a.

- **Euler–Maclaurin upgrade.** For a extending to a D_ε -differentiable f (10.5.4), §9.2's exact

remainder $R_{\{m, \varepsilon\}}$ gives an **explicit, computable** N_ε formula — replacing an existential Cauchy claim with a constructed witness. This is the concrete route by which

many closed-form series move from `+ℝ-Open` (bare existence) to `finite_diagnostic` (explicit modulus) — the "Euler–Maclaurin- based" stability condition this rung is scoped to supply.

10.5.3 Continuity algebra (`(δ_R, ε)` -continuity, closed under `+` `·` `∘`)

- **Def 10.5b.** $f = \tilde{E} \circ G_\lambda$ is ε -continuous at x with modulus η iff $\forall x'$ with $x' \sim_\eta x$

(i.e. $|x' \ominus x| \leq \eta$), $|f(x') \ominus f(x)| \leq \varepsilon$ — a finite, decidable pointwise bound, no limit invoked.

- **Th 10.12 (algebra closure, `Th_coqc` -elig).** Explicit bound arithmetic, each step a triangle-

inequality identity in $\mathbb{Q}$ (decidable order, no LEM/choice/funext — `field_simplify`-class):

- sum: $(f+g)$ is $(\varepsilon_1+\varepsilon_2)$ -continuous at modulus $\min(\eta_1, \eta_2)$;
- product (bounded f, g): $(f \cdot g)$ is $(|f(x)|\varepsilon_2 + |g(x)|\varepsilon_1 + \varepsilon_1\varepsilon_2)$ -continuous at $\min(\eta_1, \eta_2)$;
- composition: $(g \circ f)$ is ε_2 -continuous at the modulus η_1 that f needs to land inside g 's own

(ε_2, η_2) pair — a finite chain of two decidable searches, not an infinite regress. *Proof sketch (why axiom-free-eligible):* every inequality above is a finite arithmetic fact over $\mathbb{Q}$ reached by $\ominus$ -triangle-inequality plus case-split on the (decidable) order $\leq$ — no non-constructive step, so the Coq witness needs only `Qorder` / `field_simplify`; `lra`-class tactics, no classical axiom.

- **Continuum rung.** " $\forall \varepsilon > 0 \ \exists \eta > 0$ " continuity is licensed exactly as 10.5a: `Th_coqc` -eligible **once**

$\eta(\varepsilon)$ is an admissible modulus map (available in closed form for the `D_ε` -algebra-closed primitives of Th 10.8); **+ℝ-Open in general** — unwitnessed `ε-δ` existence is the named Open item (§10.13).

10.5.4 Derivative (`D_ε` exact algebra $\vdash$ declared-stability rung)

- **Primitive (unchanged).** $D_\varepsilon f := (f[n] \ominus f[n-1]) / \varepsilon$ (causal secant, §8.1/§0.5.4).
- ****Th 10.8 (`Th_coqc` AS STATED** — the exact finite- ε algebra, zero $0(\varepsilon)$ residue; local witnesses

`formal/IDM_Calculus.v`): $D_\varepsilon(f+g) = D_\varepsilon f + D_\varepsilon g$ (`delta_sum` [Toledo: Z/M.01.v1]/ `Deps_sum` [Toledo: Z/M.04.v1]) $\cdot D_\varepsilon(c \cdot f) = c \cdot D_\varepsilon f$ (`delta_scalar` [Toledo: Z/M.02.v1]) $\cdot$ the causal $D_\varepsilon(f \cdot g)[n] = f[n+1] \cdot D_\varepsilon g[n] + g[n] \cdot D_\varepsilon f[n]$ (`delta_product` [Toledo: Z/M.03.v1]/ `Deps_product` [Toledo: Z/M.05.v1]) $\cdot$ FTCC $I_\varepsilon(D_\varepsilon f)[N] = f[N] - f[0]$ exactly (`FTCC_telescope` [Toledo: Z/M.06.v1]) $\cdot$ summation by parts**

$\Sigma(f[k+1]\Delta g[k] + g[k]\Delta f[k]) = f[N]g[N] - f[0]g[0]$ (summation_by_parts [Toledo: Z/M.09.v1]). Each discharges by field / ring / finite induction — no Reals, axiom-free (Print Assumptions = Closed under the global context). The chain rule $D_\varepsilon(g \circ f)[n] = \Delta g[f[n-1], f[n]] \cdot D_\varepsilon f[n]$ (secant slope across the actual jump) is the one still described by reference, Th_coqc-elig.

- **Def 10.5c (finite- ε differentiability, the new rung).** f is **finite- ε differentiable at x**

iff the ε -indexed family $(D_{\{\varepsilon_k\}} f(x))_k$, $\varepsilon_k \rightarrow 0$ an admissible refinement sequence, is ε' -stable in the sense of 10.5a — i.e. the difference-quotient family must itself pass the Cauchy- stability test **before** $f'(x) := \lim_{\{\varepsilon \rightarrow 0\}} D_\varepsilon f(x)$ is licensed. This is the declared-stability gate the task asks for, applied to the derivative.

- **Th 10.13.** For f built from primitives closed under Th 10.8's algebra, the modulus for Def 10.5c

is computable *from the algebra itself* (chain of secant bounds through $+$ $\cdot$ $\circ$) — **finite_diagnostic**, **Th_coqc-eligible per closed-form f** . General existence for an arbitrary admissible f — no algebraic modulus supplied — stays **+R-Open**: this is the one place, flagged already at Th 10.8, where the secant-slope reading of $D_\varepsilon(g \circ f)$ can silently diverge from the naive difference-quotient reading if resolution is not matched between f and g ; restated here explicitly as the derivative rung's Open boundary, not smoothed over.

10.5.5 Integral (I_ε exact FTCC $\vdash$ Euler–Maclaurin-certified rung)

- **Primitive (unchanged).** $I_\varepsilon f[a, b]$ = the retained aggregation / antidifference (§8.2), with

FTCC $I_\varepsilon(D_\varepsilon f)[N] = f[N] \ominus f[0]$ exact (Th 10.8) — no $0(\varepsilon)$ residue, algebraic telescoping.

- **Def 10.5d (finite- ε integrability, the new rung).** f is **finite- ε integrable on $[a, b]$**

iff the family $(I_\varepsilon f[a, b])$, indexed by an admissible shrinking ε , is Cauchy-stable (10.5a) — and that stability is **certified**, not asserted, via the §9.2 exact Euler–Maclaurin identity $I_\varepsilon f[a, b] = (\varepsilon/2)[f(b) + f(a)] + \sum_{k=1}^m (B_{2k}/(2k!)) \varepsilon^{2k} [D_{\varepsilon^{2k-1}} f(b) - D_{\varepsilon^{2k-1}} f(a)] + R_{\{m, \varepsilon\}}$, giving an **explicit computable bound** $|I_\varepsilon f - I_{\{\varepsilon/2\}} f| \leq (\text{Bernoulli-term sum}) + R_{\{m, \varepsilon\}}$ — this is the "Euler–Maclaurin/FTCC-based" stability condition the task names, made literal.

- **Th 10.14.** For f with a declared bound on $D_{\varepsilon^{2k-1}} f$ over $[a, b]$ up to order m (an

explicit, checkable **finite_diagnostic** hypothesis — a finite list of numeric bounds, not a quantifier), $R_{\{m, \varepsilon\}} \rightarrow 0$ as $\varepsilon \rightarrow 0$ is machine-checkable at each declared m :

finite_diagnostic, upgrading to **Th_coqc -eligible for polynomial/rational f** where the Bernoulli tail is exactly finite (m can be taken large enough that $R_{\{m,\varepsilon\}}=0$ identically — no genuine limit needed for this class). $\int_a^b f := \lim_{\{\varepsilon \rightarrow 0\}} I_\varepsilon f[a,b]$ is licensed **only** after this certificate is produced. **General f with unbounded higher D_ε -derivatives stays $+\mathbb{R}$ -Open** — no certificate, no license.

10.5.6 Summary (the rung, once)

Sub-rung	Finite- ε primitive	Stability witness	$\varepsilon \rightarrow 0$ readout tier
10.5.1 limit	ε -stability $N_\varepsilon(a)$ (Def 10.5a)	computable modulus $\text{map } \varepsilon \mapsto N_\varepsilon$	finite_diagnostic (witnessed) / $+\mathbb{R}$ -Open (bare $\exists$)
10.5.2 series	partial sum $S_n = I_1 a[n] +$ 10.5a	Euler–Maclaurin $R_{\{m,\varepsilon\}}$ (explicit N_ε)	finite_diagnostic
10.5.3 continuity	(ε, η) -continuity (Def 10.5b)	$\eta(\varepsilon)$ modulus, closed under $+\circ$ (Th 10.12)	Th_coqc -elig (algebra) / $+\mathbb{R}$ -Open (unwitnessed $\varepsilon \rightarrow 0$)
10.5.4 derivative	$D_\varepsilon f$ exact algebra (Th 10.8)	Def 10.5c Cauchy- stability of the D_ε family	finite_diagnostic / Th_coqc -elig (closed- form f) / $+\mathbb{R}$ -Open (general)
10.5.5 integral	$I_\varepsilon f$ exact FTCC (Th 10.8)	Def 10.5d Euler– Maclaurin remainder bound	finite_diagnostic / Th_coqc -elig (poly/rational f) / $+\mathbb{R}$ -Open (general)

Open (named, tied to §10.13): general $\varepsilon \rightarrow 0$ / Cauchy-modulus existence with **no** exhibited witness, for an unrestricted admissible f — the frontier this skeleton fences rather than hides.

10.6 Probability and measure as retained frequency (a discrete measure theory)

(New chapter-companion to the discrete calculus; Dr , LLN **Th_coqc -eligible**.)

- **measure** $\mu_\lambda := I_\varepsilon(\text{indicator})$ — a coarse-grained retained count; **probability** $p_i =$

$|\text{ampi}|^2 / \Sigma |\text{ampj}|^2$ (Born-rule, already **Thcoqc in the corpus**) is the special normalized case; ****expectation**** = retained average; ****law of large numbers**** = A8 stability of the empirical mean (the plateau of $CN = (1/N)I_\varepsilon(s)[N]$). Kolmogorov's axioms appear as theorems about admissible retained counts, not primitives. ***(Strong LLN stays $+\mathbb{R}$ -Open)***

10.7 Reconciliations with the external record (กระต่ายขอด)

- **Causal set theory** (Bombelli–Lee–Meyer–Sorkin 1987, *PRL* 59, 521; Sorkin 2003, gr-qc/0309009),

`finite_diagnostic`. **ADOPT re-adapted**: an admissible causal order $<_{\lambda}$ over $\sim_{\lambda}$ -fibers plus δ_R counting fixes the readout geometry — $\text{Vol}_{\lambda}(\text{interval}(x,y)) := \#\{z : x <_{\lambda} z <_{\lambda} y\}$ restates "order + number = geometry"; I_{ε} over the chain recovers the finite- ε line-element. **This is a finite volume-by-counting kernel and is `Th_coqc-ELIGIBLE`** — `rank` (= minimal-step), the strict-partial-order facts (irreflexivity / transitivity / asymmetry of $<_{\lambda}$), and the interval $\#\{z : 0 < z < 3\}$ counted are all finite, decidable statements over $\sim_{\lambda}$ -fibers, no continuum invoked. **Honesty note (tier correction)**: a standalone witness `.v` for this kernel — referenced in earlier drafts as `formal/InfoCausalPartialOrder_attempt.v` — is **not present** in this repository's `formal/` and is **not** in `formal/verify.sh`, so it is tiered `Th_coqc-eligible` here, NOT cited as a shipped machine-checked witness (writing that witness is a named backlog item). **Local finiteness** re-adapted: a G_{λ} -fiber between any causal pair is finite — a *structural* reason sub- λ resolution is unreachable, strengthening A8. **CONFLICT (do not blend)**: causets take discreteness as ontological substrate; we do not (discreteness is a readout fact at declared λ ; the continuum is a non-readout). **Sorkin's Hauptvermutung stays $+\mathbb{R}$ -Open** (§10.13); sprinkling is cited as a `Dr` template for the stability step, not adopted as mechanism.

- **Constructive / Bishop analysis** (Bishop 1967; Bishop–Bridges 1985; Bridges–Richman 1987; Simpson

2009). **Th 10.9 (Th_coqc)**: **cotransitivity** is the constructive substitute for trichotomy, matching A8. Bishop reals (Cauchy-with-modulus) = our $\mathbb{R}$ -as-readout (`Dr`); **located sets** $\leftrightarrow$ admissible distance maps on G_{λ} -fibers; **LPO/WLPO/LLPO** rejected — the same move as A8's ban on pre-stability limits (`finite_diagnostic`). **What we add beyond Bishop**: limits are a *derived, declared, post-stability* readout (not a definitional primitive), and a genuine discrete calculus (D_{ε} , I_{ε} , FTCC, δ_R) Bishop lacks.

- **Category / type theory** (Mac Lane 1998; Grothendieck SGA1; Jacobs 1999; Johnstone 2002;

Hofmann–Streicher 1998; HoTT 2013; Coq `Coq.Classes.Setoid`). G_{λ} is an **idempotent endofunctor / reflector** (`finite_diagnostic`); the admissible description $E = \tilde{E} \circ G_{\lambda}$ is a **comma-category / Kan-extension** factorization (`Th_coqc-elig`); sets-as- G_{λ} -fibers = a **Grothendieck fibration** (indexed-set semantics); equality $=_{\lambda}$ = a **coequalizer / smallest congruence**. **Admissible descriptions are Setoids** — the direct Coq formalization vehicle (`Th_coqc`). **CAVEAT ($+\mathbb{R}$ -Open)**: a **topos** structure is explicitly NOT claimed.

10.8 Mined confirmations from the machine-checked corpus (solver-arc-private)

Each is an existing `Th_coqc` witness that grounds a textbook claim:

- **RD1–RD9** $\rightarrow$ **D** $\cong$ $\mathbb{N}$ (Peano-compatibility) $\cdot$ **RDL** 4-valued paraconsistent core (Belnap–Dunn FDE) $\cdot$

cosmogenesis order backbone ($\delta_R \rightarrow$ asymmetry $\rightarrow$ temporal order $\rightarrow$ τ_c seed $\rightarrow$ atomicity) $\cdot$ **causal order** $\prec$ and **L_R structural facts** (bridge to the spine $K \cdot L_R$) $\cdot$ **genesis-chain integration (M2)** + **readout-lossiness (CG-05A)** $\cdot$ **coupled discrete-spine stability** on the graph carrier $\cdot$ **the info-operator keystone** $B(\Phi, \Phi) = I(\Phi)$ (Dirichlet energy = retained information — **Th 5.1**, now **Th_coqc**, local witness `formal/IDM_Keystone.v`) $\cdot$ **DEC toolkit** (discrete exterior calculus around L_R , 9 tools, minimal-cell scope) $\cdot$ **π and φ as readout-invariants** (reconstruction limits, not root objects; `Dr`). (Witnesses, sister repo `solver-arc-private/formal/`, not local to this repo: `RD.v`, `RDL.v`, `RDL_Distinguishability.v`, `RDL_CausalOrder.v`, `RDL_GenesisLink.v`, `RDL_SpineGraph*.v`, `URCF_RD_All.v`, `DEC_TOOLKIT.md`.)

10.10 Geometry from the causal graph (network/graph theory + the QG export, reconciled)

(From the resume run: *network/spectral-graph reconciliation + the causal-quantum-gravity spine export.*)

- **Resistance distance = our distance** (**Th 10.15**, **Th_coqc**). $d_\lambda(i, j) := (e_i - e_j)^\top L_R^+ (e_i - e_j) \in \mathbb{Q}$

(L_R^+ the pseudoinverse) — the graph resistance distance IS the retained-resistance metric of §4.1, now a *forced rational geometry* of the causal graph. Distance stops being primitive. (Klein–Randić 1993; Doyle–Snell 1984.)

- **L_R structural facts to REUSE (Th_coqc)**. L_R rational, symmetric, PSD; ****nullity** = number of

connected components (**lap_kills_constants generalized**); Kirchhoff / matrix-tree** (spanning-tree count = any cofactor); L_{sym} , L_{rw} are rescalings of the same operator. (Spectral/algebraic graph theory: Chung 1997; Godsil–Royle 2001.)

- **Random walk / heat kernel** = $D_\epsilon / I_\epsilon / \text{FTCC (finite_diagnostic)}$. the discrete heat step

($1 - \tau L_R$) and the random walk on the graph are instances of the causal calculus (Part VIII) — the transition operator is I_ϵ of the Laplacian flow; no new machinery.

- **Discrete curvature = second δ_R** (combinatorial defect-sum **Th_coqc**-elig; the $2\pi/\text{radian-angle}$ Gauss–Bonnet form is a **+R-Open** continuum readout, §4.4). curvature is the second retained

difference; a **discrete Gauss–Bonnet** and a **discrete Bianchi** (abelian, from commuting finite differences) hold; **Forman–Ricci** $F(e)=4-\deg(u)-\deg(v)$ is native graph curvature with zero continuum limit — offered *after the source repo's honest retraction of eight attempts to force a Schwarzschild curvature onto L_R* (a worked non-readout-discipline case study). (Forman 2003; Bombelli et al. for the order-signature.)

- **Discrete Lorentzian signature + boost-invariant box symbol (Th_coqc)**. a sign $(-1$ timelike / $+1$

spacelike) on each causal edge gives a causal bilinear form invariant under order-preserving relabelling ("discrete boosts"): $\text{box_quad}(a_{tt}, a_{xx}) = -2a_{tt} + 2a_{xx}$ transforms exactly as $(g^2(1-v^2)) \cdot \text{box_quad}$ ($=1$ under the Minkowski constraint) — self-adjointness, Euclidean reduction, and boost-invariance all $\text{Th_coqc}/\mathbb{Q}$. **Fences:** box =dispersion is a *bounded restatement* (M,K posited), NOT a one-root derivation; the continuum $\square$ -limit is Open ; QNM $\sim 0.1\text{--}1.2\%$ match is finite_diagnostic only (transcendental target = non-readout). (*cqg spine export*.)

- **Spectral ceiling / τ_c floor (Th_coqc)**. a Rayleigh-quotient bound gives the spectral ceiling

without any eigenvalue-existence theory — the exact λ_2 (Cheeger/Fiedler) as a real number stays $+\mathbb{R}\text{-Open}$; use Sturm brackets for a $\mathbb{Q}$ -enclosure instead.

10.11 Mined from readoutuniverse and readoutgenesis (logic dynamics + root axioms)

- **Logic as residual flow (finite_diagnostic , readout_universe C1–C3)**. inference is a ****monotone**

descent: **under constant evidence a Lyapunov functional V decreases to 0 (C1: $V \rightarrow 8.2e-26$, 0 violations); a settled conclusion revises**** under an evidence flip (C2: $+1 \rightarrow -1$). So *proof itself is a discrete gradient flow of retained distinction* — the dynamical face of Part I's logic.

- ****Vagueness / Sorites**, machine-checked axiom-free (Th_coqc , `code/UPL_Sorites.v` , readout_universe

C7).** the sorites is resolved on the monotone scope without a sharp cutoff — reuse this witness for the book's treatment of a "gap $\neq$ assertion" (Ax-RDL4): a heap-predicate is a coarse-grained readout, not a knife-edge.

- ****The nine root axioms RD1–RD9** ("the information DNA", *readoutgenesis PART I / readoutuniverse*

v2.0).** the exact same RD1–RD9 that generate D (Part II) are the shared root of both repos — confirming the number tower's axioms are the canonical foundation, not this book's invention.

- **Geometry-dominance defect bound (`finite_diagnostic` , `readout_genesis`).** the living-geometry mixing

current (II.8a) carries a *computable* defect bound (per-operator boundary `h_exact≈0.6827`) — a concrete finite readout of "how much geometry moved", usable in Part IV/X's curvature layer.

10.12 Reconciliation with Shannon information theory (กระต่ายขูด)

(Shannon 1948; Cover–Thomas 2006; Kolmogorov 1965; Chaitin 1966.)

- **Entropy = expected retained-distinction count (`Dr` / `Th_coqc-elig`).** $H = \sum p_i \cdot \log_2(1/p_i)$

is the *expected number of binary distinctions (bits)* to name one retained state — literally the expectation (§10.6) of the `log` operator (§7.3). Information "is" retained distinction; entropy is its retained average. **Native, not imported.**

- **Channel = coarse-graining `G_λ`.** a noisy channel is exactly a `G_λ` that discards sub-resolution

distinctions; capacity = the most retained distinction that survives it.

- **Data-processing inequality = coarse-graining monotonicity (`Dr` ; the entropic `log` -form is `+ℝ-Open` — `log` over $\mathbb{Q} \rightarrow \mathbb{R}$ needs the continuum).** post-processing

(another admissible map through `G_λ`) can never *increase* retained distinguishability: $I(X;Z) \leq I(X;Y)$ for $X \rightarrow Y \rightarrow Z$ — a one-line consequence of $E = \tilde{E} \circ G_\lambda$ (composition of coarse-grainings loses, never gains). This makes the DPI a theorem about admissible descriptions.

- **Kolmogorov complexity = minimal generating record.** the complexity of a readout = the length of

the shortest `δ_R`-record whose self-composition ($\hat{}$, §7.3) regenerates it — the same "generator readout" as $\sqrt{} / \log$, now on programs. Randomness = a record with no shorter generator (a non-compressible retained distinction).

10.13 The honest Open frontier (`+ℝ-Open` / `Dr` — named, not hidden)

Continuum limit $L_R \rightarrow \square$ (d'Alembertian) · a full **topos** completeness of the substrate · **Hauptvermutung** (uniqueness of a faithful causet embedding) · the **strong** law of large numbers · π / ϕ as exact objects (they stay readout-invariants) · general $\varepsilon\text{--}\delta$ existence without a stability witness. These are the declared boundary; the book states them as Open and predicts the readout, per Part 0.

10.14 The named-Open-problems frontier, in depth ($+\mathbb{R}$ -Open — declared, not dissolved)

§10.13 named four items as the declared boundary in one line each. This section expands each to the depth the rest of the book uses elsewhere: the mainstream statement, our readout-first stance, and an explicit honesty fence. **None of the four is closed here.** Where earlier parts (§4.4, §10.6, §10.7, Part XXI) already carved a genuinely finite kernel out of a classical question, that kernel is cited as what it is — `Th_coqc` or `finite_diagnostic` — and the remaining classical question is still named $+\mathbb{R}$ -Open. Two of the four (topos completeness, the strong LLN) are declared Open outright, with no partial dissolution claimed; the other two (Hauptvermutung, exact π/ϕ) split into a closed finite kernel plus a genuinely $+\mathbb{R}$ -Open remainder — that split is stated plainly, not blurred into a single verdict. (*Diagnoses `Dr`; no new `Th_coqc` theorem is introduced in this section.*)

10.14.1 Hauptvermutung — the "Main Conjecture" of triangulation uniqueness

- **Mainstream statement(s).** Two distinct questions travel under this name and must not be conflated:

(a) the classical PL Hauptvermutung (Steinitz/Tietze, early 1900s): are any two triangulations of the same topological space always related by a common subdivision (PL-equivalent)? **Answered — and refuted** — by Milnor (1961, Reidemeister-torsion counterexample) for general polyhedra/manifolds of dimension ≥ 5 ; it holds in dimensions ≤ 3 and is subtle in dimension 4 (Kirby–Siebenmann obstruction). This part of the question is *closed by the external record*, not open — and closed in the negative in general. (b) **Sorkin's causal-set Hauptvermutung** (Bombelli–Lee–Meyer–Sorkin 1987; Sorkin 2003, gr-qc/0309009), the sense this book actually inherits (§10.7): does a causal set that is "faithfully embeddable" (via Poisson sprinkling) into a Lorentzian manifold have that manifold **uniquely**, up to approximation at the sprinkling density — or could two macroscopically different continuum manifolds both faithfully embed the same causet? This is a genuinely **open conjecture** in the causal-set literature, not resolved either way.

- **Our readout-first stance.** We inherit only sense (b), and only partially reframe it. §10.7 already

states the reconciliation: an admissible causal order $\leq_\lambda$ plus $\delta_\mathbb{R}$ counting **fixes the readout geometry** at declared λ — `Vol λ (interval(x, y))) := #{z : $x \leq_\lambda z \leq_\lambda y$ } is a finite, Th_coqc identity — a finite volume-by-counting kernel, Th_coqc-eligible (§10.7; no standalone witness .v is in this repo's formal/ — see the §10.7 honesty note). That is the finite kernel, and it is closed. What stays open is the converse uniqueness claim about the completed continuum manifold — that a faithful embedding, if one exists, is essentially the only one. This is not a question our framework can dissolve by refusing an injected non-readout, the way Part XXI dissolves Banach–Tarski or Zeno: the question is not "does an I1–I4 infinity get reached," it is a genuine combinatorial-uniqueness`

question about whether the finite readout data (the causet) underdetermines the continuum object it is sprinkled into. Declining to reify the completed manifold (§10.13) does not settle *how many* completed manifolds would be compatible with a given finite causet if one *were* formed — that is exactly the un-asked question our stance declines to answer, in either direction.

- **Honest fence.** Dr (finite volume-by-counting kernel, Th_coqc -eligible per §10.7, no standalone

witness .v in-tree) + +R-Open (uniqueness of the embedding is a named, un-dissolved conjecture — §10.7, §10.13). We claim **no** theorem, in either direction, about faithful-embedding uniqueness. Sense (a) above (classical PL Hauptvermutung) is outside this book's scope entirely — it is a fact about the external mathematical record (refuted in general), not a claim this framework makes or needs.

10.14.2 The strong Law of Large Numbers

- **Mainstream statement.** For i.i.d. $X_1, X_2, \dots$ with finite mean μ , the **strong** LLN (Kolmogorov)

asserts almost-sure convergence: $P(\lim_{n \rightarrow \infty} (1/n) \sum_{k \leq n} X_k = \mu) = 1$ — a statement about the behavior of a **completed infinite sequence**, quantified over an uncountable sample space, with convergence holding outside a measure-zero exceptional set that is itself not finitely exhibited.

- **Our readout-first stance.** §10.6 already gives the finite kernel: $\text{measure} := I_\varepsilon(\text{indicator})$,

probability the normalized case, and **the law of large numbers = A8 stability of the empirical mean** — the plateau of $C_N = (1/N) \cdot I_\varepsilon(s)[N]$ as N grows through an admissible declared sequence, with a computable stabilization witness. That is the **weak-LLN-shaped, finite- ε content**, and per §10.6 it is Th_coqc -eligible: a genuinely finite, checkable stability certificate, not a limit taken as real. The **strong** form is a strictly different claim — it is not "the empirical mean stabilizes to declared tolerance as you extend the run," it is "convergence holds for *every* infinite continuation, with probability exactly 1 over the completed infinite product space." That quantifies over the **completed infinite sequence** (I2, Part 0) and an **uncountable sample space** (§10.2's potential-vs-actual-infinite boundary) simultaneously — two separate non-readouts stacked. We do not have, and do not claim, a finite- ε substitute for "holds for all infinite continuations simultaneously": any finite run only ever exhibits *one* prefix of *one* continuation. This is not a case where refusing the injection dissolves the paradox and the maya still computes the *same* claim (contrast Part XXI) — the strong form's content is specifically the almost-everywhere behavior of the full infinite ensemble, which has no finite-readout restatement we have found. **We state this plainly as declared Open, not dissolved.**

- **Honest fence.** Th_coqc -eligible (the finite A8-stability/weak-shaped kernel, §10.6) + +R-Open

(the strong, almost-sure-over-the-completed-infinite-ensemble form, §10.13). No claim is made that the strong LLN is false, dissolved, or unnecessary for classical probability — only that this framework does not currently possess, and does not manufacture, a readout-first substitute for it.

10.14.3 A full topos completeness of the substrate

- **Mainstream statement.** An **elementary topos** is a category with finite limits, exponentials, and

a subobject classifier Ω — internal logic (typically intuitionistic, classical under extra axioms), internal set theory, and a genuine internal notion of "truth value" object. Claiming the admissible- description substrate $(G_\lambda, \tilde{E})$ (§0.5) is a topos would license the **full internal-logic and power-object machinery** of topos theory on top of it.

- **Our readout-first stance.** §10.7 already states, precisely, how much structure we have earned and

where we stop: G_λ is an idempotent endofunctor / reflector (`finite_diagnostic`); the admissible factorization $E = \tilde{E} \circ G_\lambda$ is a comma-category / Kan-extension pattern (`Th_coqc`-elig); sets-as- G_λ -fibers form a Grothendieck fibration (indexed-set semantics); admissible descriptions are Setoids — the direct Coq formalization vehicle (`Th_coqc`). These are genuine, checkable pieces of categorical structure. **What is explicitly not claimed, and is not being claimed here either, is the subobject classifier and the full topos axioms** — we have not exhibited an Ω -object for the admissible- description category, nor verified the topos axioms (pullback-stable classification of all subobjects, cartesian closure with exponentials for arbitrary admissible objects) against G_λ -fibers at every declared λ simultaneously. Constructing that classifier would very likely itself require forming a completed object across all λ (or an internal notion of "all admissible subsets," which is exactly the §10.2 potential-vs-actual-infinite boundary) — so this is not a case of "we haven't gotten to it yet," it plausibly sits on the same non-readout boundary as the rest of §10.13, though we do not claim to have proven that connection either — the honest state is simply: **not attempted, not claimed.**

- **Honest fence.** `finite_diagnostic` / `Th_coqc`-elig for the partial categorical structure actually

exhibited (reflector, fibration, Setoid formalization, §10.7) + `+R-Open`, declared outright, for full topos completeness (subobject classifier, internal logic). This is the most straightforwardly Open of the four items here: no partial dissolution is claimed for the classifier itself, only for the structure around it.

10.14.4 Exact π / φ objects

- **Mainstream statement.** Classically, π and φ are specific real numbers — elements of the

Dedekind-complete ordered field $\mathbb{R}$ — each with a unique, fully determined (if uncomputable-in-full) infinite decimal expansion. "Exact" here means: a single, completed point of the continuum, not merely an algorithm that approximates it to any finite precision.

- **Our readout-first stance — this is the sharpest split of the four, and §4.4 already draws it.**

φ is **not** an open item for us at all in its algebraic form. It is the Perron–Frobenius eigenvalue of the rational matrix $K_F = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$, the positive root of $\lambda^2 - \lambda - 1 = 0$ — an algebraic number given exactly by its minimal polynomial over $\mathbb{Q}$, with exact rational arithmetic on the quadratic-extension representation $a + b\sqrt{5}$ ($a, b \in \mathbb{Q}$). Every operation and comparison on φ in this form is decidable over $\mathbb{Q}$ (Th 4.5, Th_coqc, algebraically exact) — no continuum, no infinite decimal, no non-readout is required to pin φ down exactly. π splits into the two routes §4.4 already names: route (i), the finite order-reversal phase ($r = -1 = e^{i\pi}$) as the exchange factor of an involutive adjacent order-reversal, with the odd- k witness `formal/IDM_Harvest.v`), is Th_coqc-eligible and continuum-free — it is a finite, algebraic fact about admissible order-reversal, not π -the-real-number. Route (ii), the metric-calibration circumference/diameter ratio on the topological circle $\theta \in \mathbb{R}/\mathbb{T}\mathbb{Z}$, **uses the completed real circle** and is therefore $+\mathbb{R}$ -Open, not Th_coqc (§4.4, §10.13). **So the genuinely Open content is narrower than "is π/φ exact"**: for φ there is no Open remainder in the algebraic reading; for π the Open remainder is specifically route (ii), the continuum-metric embedding, not the order-reversal root. What is Open for *both*, strictly, is treating either constant as a **completed point of $\mathbb{R}$ -as-Dedekind-cut** rather than as a readout-invariant of a quotient grammar (§4.4) or a Bishop-style regular-Cauchy-sequence readout (§3.4) — that completed-continuum-point reading is exactly the non-readout §0.2/§10.13 declines to form, for π and φ alike, and for every other real number on the same footing.

- **Honest fence.** φ : Th_coqc (algebraically exact, no Open remainder in the quadratic-extension

reading). π : Th_coqc-elig (route (i), order-reversal) + $+\mathbb{R}$ -Open (route (ii), metric-circle calibration — §4.4, §10.13). Both: $+\mathbb{R}$ -Open if the question is reread as "is this a completed point of the Dedekind-complete continuum," since forming that completed continuum is declined by choice (§0.2, §10.13, Part XXI), not achieved and then hidden.

10.14.5 Summary table

item	mainstream question	finite kernel we hold (Th_coqc / finite_diagnostic)	what stays $+\mathbb{R}$ -Open
Hauptvermutung (causet sense)	is a faithful causet $\rightarrow$ manifold embedding unique?	causal-order volume-by-counting, Vol $_{\lambda}$ (§10.7, Th_coqc-eligible)	uniqueness of the embedding itself — un-dissolved conjecture

item	mainstream question	finite kernel we hold (Th_coqc / finite_diagnostic)	what stays $+\mathbb{R}$ -Open
strong LLN	a.s. convergence over the completed infinite ensemble	A8-stability plateau of the empirical mean, finite- $\mathbb{N}$ (§10.6)	the completed-infinite-sequence, almost-everywhere claim — no finite-readout restatement found
topos completeness	does the substrate form an elementary topos (Ω , full internal logic)?	reflector / Kan-extension / fibration / Setoid structure (§10.7)	the subobject classifier and full topos axioms — not attempted
exact π / φ	are they completed points of $\mathbb{R}$?	φ fully (Th_coqc , algebraic); π route (i) only (order-reversal, Th_coqc -elig)	π route (ii) (metric-circle calibration); both, reread as Dedekind-complete-continuum points

The discipline this section exists to model: naming an item in §10.13's one-line list is not license to leave it unexamined. Each of the four, examined, either stays flatly Open (strong LLN, topos completeness — no partial dissolution claimed) or splits cleanly into a closed finite kernel and a named Open remainder (Hauptvermutung, π / φ) — and the split is stated, not smoothed into either "we solved it" or "it's all open." Nothing here is claimed as a new Th_coqc theorem; §10.14.1–10.14.4 are Dr -tier analysis over already-cited results, sharpening the fence, not moving it.

Part XI — Closing the continuum: nothing computed by "infinity" is left standing

Parts VI and VIII established *that* the continuum is a readout; this Part discharges the promise operationally. For every operation classical analysis defines through a completed limit — derivative, integral, limit, differential equation, special function — we give a **finite- ε discrete method that returns the same benchmark answer without the continuum as a primitive**. The $\varepsilon \rightarrow 0$ statement stays $+\mathbb{R}$ -Open (A8); the *readout* — the number — is reproduced at finite ε . **Global result: 100/100 world-benchmark continuous problems reproduced** (Appendix E). No computation is left that requires an actual infinity.

11.1 The discrete derivative (replaces the difference-quotient limit)

$D_\varepsilon f[n] = (f[n] - f[n-1])/\varepsilon$ (causal/backward, error $0(\varepsilon)$) or the central $(f[n+1] - f[n-1])/2\varepsilon$ (error $0(\varepsilon^2)$); the $\varepsilon \rightarrow 0$ primitive derivative is refused, and **Richardson extrapolation** removes the leading error term — matched to the scheme's order: $(4 \cdot D_{\varepsilon/2} - D_\varepsilon)/3$ for the **central** difference (it kills the $0(\varepsilon^2)$ term, lifting the readout to $0(\varepsilon^4)$), or $2 \cdot D_{\varepsilon/2} - D_\varepsilon$ for the **backward** difference (it kills the $0(\varepsilon)$ term). Using the $4/3$ form on a backward difference would leave an $0(\varepsilon)$ residue — the extrapolation weights must match the difference scheme's leading order — so the finite readout matches f' to machine precision only when the two are paired correctly. The exact algebraic rules (Th_coqc, §8/§10.5) are the *native* content: $D_\varepsilon(f+g) = D_\varepsilon f + D_\varepsilon g$, $D_\varepsilon(fg)[n] = f[n]D_\varepsilon g + g[n-1]D_\varepsilon f$, chain via the secant slope — **zero $0(\varepsilon)$ residue**, no continuum. (Validated: 16/16 derivatives, Appendix E.B.)

11.2 The discrete integral (replaces the Riemann/Lebesgue continuum integral)

Aggregation $I_\varepsilon =$ prefix sum; the composite trapezoid $I_\varepsilon f = \varepsilon(\frac{1}{2}f_0 + f_1 + \dots + \frac{1}{2}f_N)$ with the first **Euler–Maclaurin** endpoint correction $-\varepsilon^2/12 \cdot (f'_N - f'_a)$ (§9.2) reproduces $\int$ to high order. **FTCC** $I_\varepsilon D_\varepsilon f = f[N] - f[0]$ is exact by telescoping (Th_coqc). Improper/oscillatory integrals are closed by **admissible reparametrization** ($\int_0^1 \ln(1/x) dx$ via $x=e^{-t}$ $\rightarrow \int_0^\infty t e^{-t} dt$; $\Gamma(\frac{1}{2})$ via $x=t^2$ $\rightarrow 2\int_0^\infty e^{-t^2} dt$) and by **half-period summation** + **Euler acceleration** for conditionally convergent tails (Dirichlet $\int_0^\infty \sin x/x = \pi/2$). (Each ∞ bound is a **finite truncation to a declared cutoff** plus the E–M/acceleration tail — *finite_diagnostic*; the completed $\int_0^\infty$ stays $+\mathbb{R}$ -Open.) (Validated: 34/34 integrals, Appendix E.A.)

11.3 The discrete limit (replaces the ε – δ limit taken as a primitive)

A limit is a **declared, post-stability readout**, never a definitional given (A8). The finite- ε method evaluates the sequence at $n = M \cdot 2^j$ and applies **Richardson extrapolation** on

$h=1/n$, killing each term of the asymptotic expansion in $1/n$ successively — so $(1+1/n)^n \rightarrow e$, $H_n - \ln n \rightarrow \gamma$, $n(e^{1/n}-1) \rightarrow 1$, $(n!)^{1/n}/n \rightarrow 1/e$ are all recovered to $1e-6+$ from purely finite data. The value is the **A8 plateau**, tagged `finite_diagnostic`; the completed $\varepsilon \rightarrow 0$ object stays `+R-Open`. (Validated: 15/15 limits, Appendix E.C.)

11.4 The discrete ODE (replaces the continuous initial-value problem)

$y'=f(x,y)$ is the difference equation $y[n+1]=y[n]+I_\varepsilon(f)$ — the continuous ODE was only ever a stability limit of this. RK4 is the fourth-order I_ε quadrature of the vector field across one step; run to the endpoint it reproduces the named closed form to the declared tolerance ($y'=y \Rightarrow e$, $y'=2xy \Rightarrow e^{x^2}$, $y'=1/(1+x^2) \Rightarrow \arctan$, Riccati $y'=y^2 \Rightarrow 1/(1-x)$). No continuum flow is invoked — only accumulated retained steps. (The RK4 readout at a declared step count is `finite_diagnostic` over $\mathbb{Q}$; the named completed value — `e`, `arctan x`, ... as reals — stays `+R-Open`.) (Validated: 12/12 ODEs, Appendix E.D.)

11.5 Discrete special functions (replaces continuum special functions)

Each "transcendental" function is a **convergent discrete process** — a series, a quadrature, or a continued fraction — never a continuum primitive: Γ = discrete quadrature of $t^{s-1}e^{-t}$ (or the factorial recurrence); $\zeta(s)$ = Euler–Maclaurin-accelerated partial sum (giving $\zeta(2)=\pi^2/6$, $\zeta(4)=\pi^4/90$, $\zeta(3)$ Apéry, $\zeta(1/2)$ regularized); `erf` = its Maclaurin series; `Catalan`, `Li2`, elliptic `K`, Bessel `J0` = their defining discrete sums/products; the Wallis product and Dirichlet $\text{Si}(\infty)=\pi/2$ from finite acceleration. (Each value is that discrete process summed/quadratured to a declared tolerance — `finite_diagnostic/exact` over $\mathbb{Q}$; the named completed constant — $\pi^2/6$, Apéry $\zeta(3)$, $\pi/2$, ... as reals — stays `+R-Open`.) (Validated: 23/23 special-function values, Appendix E.E.)

11.6 Closure theorem (the continuum is dispensable)

Closure (Dr, computationally witnessed). Every continuum computation tested — derivative, integral, limit, ODE, special-function value — is reproduced by a finite- ε discrete method to the declared tolerance, using only $\mathbb{Q}$ -data, `D_ε`, `I_ε`, admissible reparametrization, and A8-declared acceleration. The continuum enters nowhere as a primitive; it is recovered everywhere as a readout. Witness: **100/100**, `validation/hundred_continuum_problems.py` (Appendix E).

This does **not** claim the classical $\varepsilon \rightarrow 0$ existence theorems (they stay `+R-Open`, §10.13); it claims the strictly stronger operational fact that *the answers those theorems name are obtained without the limit being taken as real*. The honest fence: the readout is reproduced; the completed continuum is still declared Open and predicted, per Part 0.

Part XII — Abstract algebra as retained symmetry (group · ring · field)

Algebra is not a theory of abstract symbols floating above nothing; in this foundation it is the **bookkeeping of the relabelings that preserve every readout**. Structure comes first (δ_R , the operators of Part VII), and algebra is the study of its symmetries and closures.

12.1 A group is the automorphisms of a retained structure

For a retained structure F with reader 0 , define the **admissible automorphisms** $\text{Aut}(F, 0) := \{ g : g \text{ relabels } F \text{ and } 0(g \cdot X) = 0(X) \text{ for all readouts } \}$ — the reversible relabelings that change no distinction any reader can detect (gauge-type redundancies, §III of the π/ϕ synthesis).

- **Group axioms are theorems, not decrees.** Composition of readout-preserving relabelings is

readout-preserving (closure); it is associative because relabeling-composition is; the empty relabel is the identity; readout-preservation is reversible (each g has g^{-1}). **Th 12.1 (Th_coqc, witness formal/IDM_FiniteWitnesses3.v: aut_assoc / aut_id_left [Toledo: D/M.48.v1] / aut_id_right [Toledo: D/M.49.v1] / aut_inv_left [Toledo: D/M.50.v1] / aut_inv_right [Toledo: D/M.51.v1] — bijective readout-preserving relabelings are closed under $\circ$, with identity and two-sided inverse):** $(\text{Aut}(F, 0), \circ)$ satisfies the group axioms. No axiom is imported; the group is the symmetry of a reading.

- **Finite groups only, as readouts.** A group is presented by its finite Cayley table — a finite

readout. Infinite/continuous groups (Lie groups) are $+\mathbb{R}\text{-Open}$: their finite-dimensional *representations* are the readouts we admit (§12.3), not the completed manifold.

- **Symmetry = indistinguishability under a relabel** ties directly to the L_R kernel (§V): elements

a reader cannot separate sit in one orbit.

12.2 Rings and fields are the operator algebra, already grounded

The number tower is the worked example, so this chapter only makes the abstract shape explicit:

- A **ring** = a carrier with $\oplus$ (an abelian group, §7) and $\otimes$ (a monoid, §7) linked by distributivity

— all of which are the retained-operator laws of Part VII. **D is a commutative semiring** (RD1–9, Th_coqc); **$\mathbb{Z}$ is a ring** (Grothendieck completion, §III; distributivity witness formal/IDM_FiniteWitnesses3.v: ring_distrib_Z); **$\mathbb{Q}$ is a field** (field of fractions, $\div$ total except $\div 0$ refused, §7). **Th 12.2 (Th_coqc-elig, mined):** these hold with the ring/field axioms as *derived* identities of $\oplus, \otimes, \div$, not primitives.

- **$\div 0$ is the one refused operation** — a field is " $\mathbb{Q}$ minus the non-readout $1/0$ ", exactly the

partition-with-no-parts refusal (§7). The field structure is honest about its one hole.

12.3 Representations, order, and solvability

- **Representation = action on retained amplitudes.** A representation sends group elements to admissible

operators on a $\mathbb{Q}$ -space of retained states; its **dimension is an integer** (a carrier count). This is why $SU(3)$ appears with $k=3$ from the ordered-tape odd-closure witness (§II), and why the fusion dimension $FPdim(\tau)=\varphi$ is explicitly **not** an ordinary representation dimension (§4.4, Cor 4.1) — the integrality of ordinary dimensions is a Th_coqc-elig finite fact.

- **Lagrange / orbit-counting = finite counting theorems.** $|H|$ divides $|G|$; orbit size divides

$|G|$ — **Th 12.3.** The **divisibility core** is machine-checked (Th_coqc, witness formal/IDM_FiniteWitnesses2.v: lagrange_order_div: $(n/\gcd(g,n)) \mid n$); that $n/\gcd(g,n)$ is the additive order of g in $\mathbb{Z}_n$, and the general Cayley-coset partition, are standard but **not formalized here** (Th_coqc-elig / Dr). A decidable finite readout.

- **Galois solvability = radical-tower reachability (Dr).** A readout is "solvable by radicals" iff it

is reachable from $\mathbb{Q}$ by a finite tower of $\oplus, \otimes, \div, ^, \sqrt{}$ (§7 operators). The abstract correspondence (solvable group $\leftrightarrow$ radical tower) is Dr/finite_diagnostic here — stated as the framework's reading, with the general theorem over the completed field left + $\mathbb{R}$ -Open.

Part XIII — Linear algebra as the algebra of L_R

Linear algebra is not an add-on; the central object of this whole foundation — the retained-information operator L_R (Part V) — is a linear operator, and its spectral theory is where "information is the central axis" becomes computational.

13.1 Vector spaces over $\mathbb{Q}$ (finite, no smuggled completeness)

A **retained vector space** is finite $\mathbb{Q}$ -linear combinations of retained states — closure under $\oplus$ and $\mathbb{Q}$ -scaling (§7). Basis = a maximal set of mutually distinguishable states (a δ_R -independent set); **dimension** = **the number of retained distinctions** (Th_coqc -elig, a finite count). Completeness of the space (Hilbert-space limits) is **$+\mathbb{R}$ -Open** — we work in the $\mathbb{Q}$ -dense core and take the completion only as a declared readout (A8).

13.2 The retained metric, L_R , and the spectral readout

Every retained operator splits into a metric part and a load-free part (Th_coqc , witnesses `formal/IDM_Harvest.v: sym_skew_reconstruct, sympart_self_adjoint [Toledo: LR/M.17.v1], skew_diag_zero [Toledo: LR/M.19.v1]`): $M = \frac{1}{2}(M+M^T) + \frac{1}{2}(M-M^T)$ — a **self-adjoint** part (the retained metric, $= M^T$, which carries the retained load) plus a **skew** part ($= -(\cdot)^T$, whose diagonal vanishes, so it holds no retained metric on any single node). The retained metric $\langle \cdot, \cdot \rangle_G$ is exactly this self-adjoint part; L_R is already self-adjoint (§15.2 `laplacian_symmetric [Toledo: L_R/M.20.v1]`), so it is its own metric part.

- **Inner product = the positive retained metric** $\langle x, y \rangle_G = x^T G y$, $G > 0$ (§III of the π / ϕ

synthesis, Eq. 4). It makes retained load nonnegative and fixes which relabelings are admissible (the automorphisms of Part XII are the G -isometries).

- **$L_R = D_W - W$** (graph Laplacian, Part V) is the canonical self-adjoint operator. Its keystone is

$B(\Phi, \Phi) = I(\Phi)$ — the Dirichlet energy *equals* the retained information (the keystone **Th 5.1**, Th_coqc , witness `formal/IDM_Keystone.v`). Its **kernel is the indistinguishability space** (constant-on-components readouts); its **spectrum is the set of retained modes**; its **Perron–Frobenius eigenvalue** carries the dominant retained scale (the ϕ of the minimal transfer K_F , §4.4).

- **Determinant = signed retained volume; rank = retained-distinction count.** Both are exact $\mathbb{Q}$

computations (validated, Appendix D.L3); **matrix–tree** links `det` of an L_R minor to spanning trees (§15.3).

13.3 Solving linear systems (exact, no continuum)

$Ax=b$ over $\mathbb{Q}$ is solved by finite Gaussian elimination — an **exact Th_coqc-elig** procedure with no rounding and no limit. A **local discrete matrix library** ships the machine-checked core of this Part: matrix algebra $(A \cdot B)^T = B^T \cdot A^T$, $I \cdot A = A$, and the Laplacian facts of §15.2 — `formal/IDM_Matrix.v` (`transpose_mmul` [Toledo: Q/M.11.v1], `mid_left` [Toledo: Q/M.12.v1], `laplacian_symmetric` [Toledo: LR/M.20.v1], `laplacian_rowsum_zero` [Toledo: LR/M.21.v1], `laplacian_ones_in_kernel` [Toledo: L_R/M.22.v1]), all axiom-free. Eigenvalue/eigenvector computation for the finite L_R is a finite readout; the completion to **real spectra** of an infinite operator (the real spectral theorem, the continuum limit $L_R \rightarrow \square$) is $+\mathbb{R}$ -Open **by our own philosophy** — real eigenvalues need $\mathbb{R}$ -completeness (§10.13), so it was never finite "warrant" work but a declared frontier.

13.4 The averaging projector — parameter reduction by symmetry (Th_coqc)

A symmetry can *eliminate free parameters*, and the mechanism is a single linear operator: the **averaging / Reynolds projector**. For a finite symmetry group G acting on matrices, the average $\Pi(X) = (1/|G|) \cdot \sum_{g \in G} g \cdot X \cdot g^{-1}$ is an **idempotent projector** onto the G -invariant subspace, and the number of surviving free parameters equals that subspace's dimension $= (1/|G|) \cdot \sum_g \text{tr } \rho(g)$. When the only invariant is the trace, Π collapses to the **isotropy twirl** $\Pi(X) = (\text{tr } X / n) \cdot I$, a genuine **idempotent projector** — $\Pi(\Pi(X)) = \Pi(X)$ is machine-checked (Th_coqc, witness `formal/IDM_Matrix.v:twirl_idempotent`, via `trace_twirl` [Toledo: Q/M.17.v1]) — whose image is the **one-dimensional scalar line** $\mathbb{Q} \cdot I$ (`twirl_image_scalar` [Toledo: Q/M.13.v1], `scalar_line_one_dim` [Toledo: Q/M.19.v1]). So a full metric on an n -dimensional retained space — $n(n+1)/2$ free symmetric components — averaged by a symmetry whose only invariant is the trace, retains **exactly one scalar**. (*What is mechanized is the reduced twirl $\Pi(X) = (\text{tr } X/n) \cdot I$ — its idempotence and scalar image; the general finite-group average $\Pi(X) = (1/|G|) \sum_g gXg^{-1}$ and the invariant-dimension count $(1/|G|) \sum_g \text{tr } \rho(g)$ are the standard Dr reading, not separately mechanized here.*)

This is the mathematics behind a **parameter economy**: the same operator that reads out isotropy is the one that cuts the parameter count. (*It is the abstract content extracted from the Standard-Model parameter-economy program — the $B_4 = (\mathbb{Z}_2)^4 \times S_4$ signed-permutation twirl forcing an isotropic scalar metric, and the chiral projector $P_+ = \frac{1}{2}(I + \Gamma^5)$ with eigenvalues $1, 0, 1, 0$ halving the retained degrees of freedom — told here purely as the idempotent-projector / invariant-dimension theorem, tier Th_coqc for the finite scalar-image fact, Dr for the general invariant-dimension reading.*)

Part XIV — Complex analysis as the rotation-readout of retained phase

The complex numbers are introduced here **without** the mystical " $\sqrt{-1}$ ". A complex number is a retained **phase**, and **i** is the generator of a quarter-turn — an order-4 element of the rotation readout, nothing more occult than the order-reversal sign of §4.4.

14.1 $\mathbb{C}$ as $\mathbb{Q}$ -pairs with a rotation operator

$\mathbb{C}_{\mathbb{Q}} := \mathbb{Q} \times \mathbb{Q}$ with $\oplus$ componentwise and $\otimes$ the rotation-scaling $(a,b) \cdot (c,d) = (ac-bd, ad+bc)$. Then **$i = (0,1)$ is the quarter-turn**: $i^2 = (-1,0)$ because two quarter-turns are a half-turn — the same order-reversal that carries phase π (§4.4, Th 4.4). **$i = (0,1)$ is defined constructively as the positive quarter-turn, not as a square root of a negative real** — the contamination " $\sqrt{-1}$ as a real-line object" is refused. (Order 4 alone fixes only the pair $\{i, -i\}$; singling out **i** needs the orientation choice built into $(0,1)$.) The object is a finite cyclic phase. $e^{i\theta}$ is the **compact-closure readout** $U(\theta)$ with $U(\theta+2\pi) = U(\theta)$ (Eq. 11 of the π/ϕ synthesis).

14.2 Roots of unity, Euler's identity, and closure

- **Roots of unity = finite cyclic readouts.** The n -th roots are the $\mathbb{Z}_n$ phase group (validated,

Appendix D.L4; $\mathbb{Z}_3 = \{1, e^{2\pi i/3}, e^{4\pi i/3}\}$ is exactly the Standard-Model center quotient, §III of the synthesis, Eq. 14). Finite, exact.

- **Euler's identity $e^{i\pi} + 1 = 0$** is the order-reversal invariant $e^{i\pi} = -1$ (§4.4) — a

Th_{coqc} -elig algebraic fact about the compact phase, not a statement needing the real exponential's completion.

14.3 Holomorphy, contours, and residues (discrete)

- **Holomorphic = a phase-transport that closes.** A map is admissible-analytic when its transport

$U'_{\{j \leftarrow i\}} = h_j U_{\{j \leftarrow i\}} h_i^{-1}$ (Eq. 5) has trivial holonomy on contractible cycles; the **holonomy** $H_C = \prod_{e \in C} U_e$ (Eq. 6) records whether translations close consistently.

- **Contour integral = I_ϵ around a finite cycle; the winding number is a retained integer count.**

Cauchy's theorem is the closure condition (a distinction transported around a closed path returns to the same readout class $\Rightarrow$ zero net accumulation). The **winding number** $n(C, a)$ is an integer property of the cycle (a retained turning count); the **residue** $\text{Res}(f, a)$ is a *distinct* object — the Laurent coefficient a_{-1} of the function, generally not

an integer. The residue theorem multiplies them: $\oint_C f \, dz = 2\pi i \cdot \sum_k n(C, a_k) \cdot \text{Res}(f, a_k)$ — both sides finite readouts (Dr/finite_diagnostic). Full complex analysis over the completed plane (analytic continuation, the $\mathbb{R}/\mathbb{C}$ -completeness theorems) stays **+R-Open**, per Part 0.

Part XV — Combinatorics and graph theory (the native home of δ_R)

This Part is last only in order, not in fundamentality: $\delta_R = (a \# b)$ **is an edge**, so a retained structure **is a graph**, and counting distinctions is the most native computation in the whole book. Everything here sits at the native tier (`Th_coqc` -elig — finite, decidable, over $\mathbb{Q}$).

15.1 Counting is retained-distinction counting

`Counting_λ(S) = |[S]_λ|` (§10.2) — the de-duplicated count of $\sim_\lambda$ -classes, always a finite natural. Binomial/multinomial coefficients count admissible selections; **Catalan numbers, partitions, Stirling numbers, Bernoulli numbers** are all validated finite readouts (Appendix D.L3–L5). Generating functions are **readout-series** — the same finite- ε series objects of Part IX, now indexing combinatorial families.

15.2 Graphs, the Laplacian, and connectivity

A graph is the retained structure. $L_R = D_W - W$ is its Laplacian (Part V); therefore every graph-theoretic invariant is an L_R readout:

- **Connectivity** = `ker L_R`. The number of connected components equals `dim ker L_R` (constant-on-

component vectors) — the kernel is literally the indistinguishability space (§V). The **constants-in-kernel** direction is machine-checked (`Th_coqc` , witness `formal/IDM_Matrix.v: laplacian_ones_in_kernel` , via row-sum zero `laplacian_rowsum_zero` [Toledo: `L_R/M.21.v1`]); the full rank-nullity count of components over $\mathbb{Q}$ is `Th_coqc` -elig.

- **Spectral gap** = $\lambda_2(L_R)$ controls mixing/expansion — the same spectral quantity that the

particle-graph diagnosis reads (`falsify_particle_graph` , memory) and that the π / φ transfer selects (§4.4).

15.3 Trees, Euler characteristic, and finite guarantees

- **Matrix–tree theorem**. The number of spanning trees = `det` of any cofactor of L_R — a bridge to

linear algebra (§13.2), an exact $\mathbb{Q}$ determinant (`Th_coqc` -elig).

- **Euler characteristic** $\chi = V - E + F$ and the **discrete Gauss–Bonnet / Bianchi** identities (§10.10)

make curvature a second retained difference summing to a topological invariant — geometry and combinatorics meet with no continuum (Th_coqc -elig in the mined DEC toolkit).

- **Pigeonhole and Ramsey-type bounds** are finite existence readouts: with more distinctions than

classes, two must collide — a one-line δ_R counting argument (Th_coqc, witness formal/IDM_FiniteWitnesses.v: pigeonhole; the handshake identity $\sum \deg = 2|E|$ is handshake_lemma [Toledo: L_R/M.01.v1]).

Breadth closure. With Parts XII–XV the foundation now carries dedicated, tier-honest chapters for algebra, linear algebra, complex analysis, and combinatorics/graph theory — each derived from δ_R , L_R , and the Part VII operators, none importing a continuum primitive. The remaining declared frontier (topology/topos, Hauptvermutung, completed continuum limits, exact π/φ objects) stays + $\mathbb{R}$ -Open (§10.13), stated and predicted, never smuggled in.

Part XVI — Measure theory and functional analysis (the discrete floor of analysis)

Measure and function-space theory are usually built on a completed σ -algebra and a complete normed space. Here both are **readouts of the discrete**: a measure is a retained count, an integral is I_ε against it, and the operator theory is the finite spectral theory of L_R (Part XIII). Completeness is the one place the continuum enters, and it is fenced $+\mathbb{R}$ -Open.

16.1 Measure = retained count; integral = I_ε against it

- **Discrete measure.** $\mu_\lambda(S) := I_\varepsilon(1_S)$ — the retained count of admissible cells in S (§10.6).

It is **finitely additive by construction** (Th_coqc -elig): disjoint fibers add their counts. σ -additivity is the **A8-stable limit** of finite additivity — stated, and $+\mathbb{R}$ -Open for the completed σ -algebra (a countable union that "lands" is a non-readout).

- **Integral.** $\int f \, d\mu_\lambda := I_\varepsilon(f \cdot 1_S)$ — the Lebesgue integral is the readout of this finite retained

sum; the classical $\int$ is recovered by the continuum-maya bridge (Part XX target) wherever A8-stable.

- **Measurable function is replaced by the admissible map** $E = \tilde{E} \circ G_\lambda$ (§0.5) — a stipulated discrete

substitute, not proven equal to classical measurability (which is the strictly stronger σ -algebra preimage condition): here measurability is admissibility at the declared resolution; no separate σ -algebra machinery is imported.

16.2 L^p , the retained inner product, and finite functional analysis

- **L^p norm = retained- p aggregation** $\|f\|_p := (I_\varepsilon |f|^p)^{1/p}$; $\|\cdot\|_2$ comes from the positive

retained metric $\langle f, g \rangle_G = I_\varepsilon(\bar{f} g)$ (Part XIII). **Cauchy-Schwarz** $|\langle f, g \rangle| \leq \|f\|_2 \|g\|_2$ is a finite Th_coqc -elig inequality over $\mathbb{Q}$ (discrete sum form).

- **Bounded operator = finite retained coupling.** On the $\mathbb{Q}$ -space, an operator is a finite matrix;

L_R is self-adjoint (Th_coqc , witness `formal/IDM_Matrix.v: laplacian_symmetric`) and positive semidefinite (§5.1 `keystone_nonneg` [Toledo: Keystone/M.07.v1]). As in §13.3, the finite spectral content that is a *readout* is **not** Open: the **inertia** of the finite L_R (how many eigenvalues are $+/-/\emptyset$) is a **resolved $\mathbb{Q}$ sign-count** — machine-checked so a

strict resolved sign never lies, with $\perp$ reported where a pivot does not resolve (`Th_coqc`, `formal/IDM_ResolvedCount.v` / §10.2; the full classical signed-floor additionally needs Sturm backward-stability, not claimed here) — each eigenvalue admits a $\mathbb{Q}$ -enclosure via Sturm and eigenpairs to any declared resolution (`finite_diagnostic`), and rational eigenpairs where they exist are exact. What is genuinely **$+\mathbb{R}$ -Open** is narrower than "the spectral theorem": the **exact (generally irrational) eigenvalues** and the **infinite-dimensional** completion — orthogonal diagonalization over a completed $\mathbb{R}$, the retained-mode basis of an infinite Hilbert space, and its spectral measure — which need $\mathbb{R}$ -completeness (§10.13).

- **Riesz representation (finite).** Every admissible linear functional on the finite $\mathbb{Q}$ -space is

$f \mapsto \langle v, f \rangle_G$ for a unique retained v — `Th_coqc`-elig, by finite linear algebra. The infinite-dimensional Riesz theorem is **$+\mathbb{R}$ -Open**.

Part XVII — Category theory in the readout vocabulary

The substrate's universal properties are categorical, and §10.7 already identified the pieces. This Part makes them a chapter: category theory here is the theory of **admissible maps between retained structures**, with G_λ the organizing functor. It is the abstract restatement of the readout-first principle, not an external framework imported on top.

17.1 The category of retained structures

- **Category $\mathcal{R}$** . Objects = retained structures (G_λ -fibers); morphisms = admissible maps

$E = \tilde{E} \circ G_\lambda$; composition = map composition (associative); identities = the trivial relabel. A finite presentation is a finite readout (Th_coqc -elig).

- G_λ is an **idempotent reflector** (endofunctor with $G_\lambda \circ G_\lambda \cong G_\lambda$, `finite_diagnostic`, §10.7) —

coarse-graining is a categorical reflection onto the admissible subcategory.

- $E = \tilde{E} \circ G_\lambda$ reads as a **Kan-extension / comma factorization** (Dr — the factorization exists; the

universal property is stated as the framework's reading, not proved here): every admissible description factors through the reflector.

17.2 Fibrations, congruences, and the finite Yoneda reading

- **Sets-as- G_λ -fibers = a Grothendieck fibration** (indexed-set semantics); $** = _ \lambda$ = a coequalizer /

smallest congruence (§10.7) — **categorical restatements, tier Dr (no universal-property proof given here)**. Admissible descriptions are setoids $**$ — the direct Coq vehicle (Th_coqc -elig; the setoid mechanization lives in the external corpus, no local witness).

- **Functor = readout-preserving map; natural transformation = an admissible family** of such maps.
- **Finite Yoneda (the readout-first principle, categorically)**: a retained object is determined, up to

$\sim_\lambda$, by the family of its admissible readouts $\{0(g \cdot -)\}$ — "an object is its readouts" (Th_coqc for the finite readout-vector fact — witness `formal/IDM_FiniteWitnesses.v:finite_yoneda`; the full enriched Yoneda is Dr).

- **+ $\mathbb{R}$ -Open**: a **topos** structure on the substrate is explicitly **not** claimed (§10.7 caveat).

Part XVIII — Statistics and inference (bounded-reader inference, fail-closed)

Above probability (§10.6) sits inference: estimating and testing from finite retained data. In this framework inference is done by a **bounded reader** who returns a **verdict**, never an omniscient pronouncement on a "true parameter" — the reader/resolution discipline of Part 0.5, applied here as a **bounded-judge law** for statistics (every tester is itself a bounded reader), unified with the numeric-honesty discipline (`idm_discipline`).

18.1 Estimation as retained-frequency readout

- **Estimator = retained readout.** The sample mean is $\hat{C}_N = I_\varepsilon(\text{sample})/N$; consistency is the **A8**

plateau of $\hat{C}_N$ (the LLN of §10.6). A confidence interval is a declared-resolution interval** around the retained estimate — its width is the declared ε , not a claim about a completed sampling distribution.

- **Bayesian update = retained reweighting.** Posterior $\propto$ prior $\times$ likelihood is a reweighting of

retained-distinction mass; the **Born-rule normalization** $p_i = |a_i|^2 / \sum |a_j|^2$ (§10.6, `Th_coqc` in the corpus; Part V Pr 5.1) is the special normalized case. Tier `finite_diagnostic / Dr`.

18.2 A hypothesis test IS a verdict

A test returns a **Verdict(ACCEPT/HOLD/BLOCK)** (`idm_discipline`), fail-closed:

- **ACCEPT** — the effect reads **distinct at the declared resolution** ($\neg \text{eq_eps}$ on the retained

contrast); **HOLD** — underpowered / budget-exhausted / assumptions unread (never a silent "no effect"); **BLOCK** — a modelling assumption is violated (e.g. a sorites/non-transitivity trap in the comparison chain, `eq_chain_guard`). The **p-value** is a declared-resolution overlap fraction, not a gate to be crossed. This eliminates the silent-wrong-answer class (a **HOLD** is not a **NULL**).

- **Bounded-judge law.** The tester is itself a bounded reader; "significance" is a readout at declared

resolution, never a truth certificate (`Dr`).

Part XIX — Optimization (retained stationarity, obstruction-zeroing)

Optimization is finding where a retained cost stops changing, at a declared resolution. Its objects are the tools this book already ships: the gradient is D_ε , the search is I_ε of the descent field, and the stopping test is a fail-closed verdict.

19.1 Gradient, convexity, stationarity

- **Gradient** = D_ε . The objective J is a retained cost readout; its gradient is the finite difference $D_\varepsilon J$ (exact in the causal form, §8). **Stationarity** is $D_\varepsilon J = 0$ at the declared resolution — solved by **obstruction-zeroing** (`solve_obstruction`, `idm_discipline`), which returns `ACCEPT` only when the residual reads zero and `HOLD` otherwise.

- **Convexity** = **nonnegative retained second difference**. $D_\varepsilon^2 J \geq 0$ (a decidable $\mathbb{Q}$ sign check)

certifies that **any retained stationary point is a global minimum, if one exists** (`Th_coqc`-elig); it does **not** give existence or uniqueness on its own ($J=x$ is convex with no minimum; $J=\text{const}$ has every point a minimizer). **Uniqueness needs strict convexity** $D_\varepsilon^2 J > 0$ (plus boundedness/ coercivity for existence). Convexity is a finite readout, not a statement about a smooth Hessian over $\mathbb{R}$.

19.2 Constrained optimization and linear programming

- **Lagrange multipliers** = **retained stationarity on the admissible fiber**. Constrained optimality is

$D_\varepsilon J = \sum \lambda_k D_\varepsilon g_k$ on the G_λ -fiber cut out by the constraints; the KKT feasibility / complementary-slackness check is `Th_coqc`-elig (a finite $\mathbb{Q}$ verdict); its equivalence to the classical KKT theorem is `Dr`.

- **Linear programming** = **exact $\mathbb{Q}$ vertex enumeration**. The optimum sits at a polytope vertex; simplex

is an **admissible path on the polytope graph** (ties to `L_R/graph`, Part XV) with exact $\mathbb{Q}$ pivots — `Th_coqc`-elig, no floating error.

- **Gradient descent** = I_ε of the **negative-gradient field** (an ODE, §11.4); convergence is the `**A8`

`plateau**`, reported as a fail-closed verdict (`HOLD` if the budget ends before the plateau). Continuous convex analysis over $\mathbb{R}$ (subdifferentials, completed minimizers) stays `+R-Open`.

(Validated: `validation/breadth2_problems.py`, 34/34 — Cauchy–Schwarz, finite spectral theorem, Riesz, G_λ idempotence, finite Yoneda, LLN plateau, Bayesian/Born normalization, test-verdict ACCEPT/HOLD, sorites BLOCK, convexity, obstruction-solve, exact- $\mathbb{Q}$ LP, gradient descent, Lagrange.)

Breadth closure (extended). With Parts XVI–XIX the foundation adds measure/functional analysis, category theory, statistics/inference, and optimization — each derived from δ_R , L_R , $I_\varepsilon/D_\varepsilon$, and the verdict discipline, none importing a continuum primitive. The completed-continuum pieces (σ -algebra limits, infinite Hilbert spaces, topos structure, continuous convex analysis) are named $+\mathbb{R}\text{-Open}$ (§10.13), and the **continuum-maya bridge** (Part XX, backlog) is the capstone that will construct the continuum layer *as a readout* and prove it computes identically.

Part XX — The continuum-maya bridge (the continuum constructed as a readout, and computed with)

The book's central claim — *the continuum is a readout of the discrete* — is here made a **constructive, testable bridge**, not merely a stance. We build the continuum layer explicitly *from* the discrete, as an **appearance** (a readout), and show it computes the classical continuum answers identically wherever those answers exist.

20.1 The construction Λ : discrete data $\rightarrow$ continuum-appearance

Define the bridge map $\Lambda : (\text{finite-}\varepsilon \text{ discrete computation}) \rightarrow (\text{continuum-appearance value})$ as the **A8-stable readout** of the finite- ε data — the plateau reached by the declared accelerator (`limit_eps` / Euler–Maclaurin / Richardson, Part XI, `tools/idm_tools.py`). Λ is **total on A8-stable inputs** and **refuses — returns HOLD** (a `Verdict`, `idm_discipline`) where no plateau exists. So the continuum Λ builds is exactly the **computable appearance**; it never forms a completed non-readout.

20.2 Faithfulness: the constructed continuum computes identically

- **Exact core** (`Th_coqc`, witness `formal/IDM_Bridge.v`). For the fundamental operation — the

Fundamental Theorem — the bridge is **exact, zero residue**: $I_\varepsilon(D_\varepsilon f) = f[N] - f[0]$ (FTCC, `FTCC_exact` [Toledo: R/M.09.v1] / `FTCC_eps_exact` [Toledo: R/M.10.v1], proved axiom-free over $\mathbb{Q}$; the ε cancels for every declared $\varepsilon \neq 0$). The continuum's FTC is *recovered as a finite telescoping readout* — no limit is taken.

- **Numeric faithfulness** (`finite_diagnostic`). For derivative, integral, limit, ODE, and

special-function operations, Λ reproduces the classical continuum value on every A8-stable case — **witnessed at 100/100** (Appendix E) and 34/34 (Appendix, Parts XVI–XIX).

- **Faithfulness statement** (`Dr`, **computationally witnessed**). $^*\Lambda(\text{discrete op}) = \text{classical continuum}$

`op` wherever the classical operation exists*, and the set where Λ refuses (`HOLD`) is **exactly** the `+ℝ-Open` non-readouts (§10.13). This upgrades the empirical 100/100 into a stated bridge property; the exact rung is machine-checked, the numeric rung is validated, and no completed continuum is ever formed.

20.3 The maya clause (honesty)

The constructed continuum is a **readout of the discrete** (conventional truth, §0.3) — an appearance, never the ultimate object. The bridge is **one-way faithful** (discrete $\rightarrow$

appearance): it *predicts* the readout on the Open frontier rather than closing it. Nothing here claims the classical $\varepsilon \rightarrow 0$ existence theorems; it claims the strictly operational fact that **the answers those theorems name are obtained without the limit being taken as real**.

20.4 The maya made precise — definitions and the completeness axiom

- **Def 20.1 (resolution family).** A *resolution family* (E_λ) is a sequence of admissible readouts at

decreasing resolutions λ , each a **finite discrete rational** computation ($E_\lambda = \tilde{E} \circ G_\lambda$, §0.5).

- **Def 20.2 (the maya map Λ).** $\Lambda((E_\lambda)) :=$ the **A8-stable plateau** of the family — the value the

finite readouts agree on, to a declared tolerance, once refinement no longer changes the readout. Λ is total on A8-stable families and returns **HOLD** (a **Verdict**, **idm_discipline**) where no plateau exists. Λ never forms a completed limit; it reads the plateau.

- **Ax 20.1 (maya-completeness — the replacement principle, **Dr**).** *Every continuum operation used in

practice — *derivative, integral (any dimension), limit, differential equation, series, special function, regularization* — admits a **Λ -preimage**: a resolution family whose plateau equals the classical value. The continuum's computational content* factors through the discrete: **there is nothing the continuum computes that the maya does not**.

20.5 The completeness theorem — the maya is a genuine replacement

Th 20.1 (maya replacement; exact core **Th_coqc, full statement **Dr** computationally witnessed).** For the entire standard analytic toolkit, Λ reproduces the classical value from finite discrete data:

classical operation	Λ -preimage (finite)	witness
derivative f'	D_ε + Richardson	<code>IDM_Calculus.v</code> (exact rules)
integral $\int (1-D)$	I_ε + Euler–Maclaurin	<code>IDM_Bridge.v</code> FTCC (exact)
integral $\int (d-D, \text{any } d)$	radial finite readout, §20.6	<code>infinity_accuracy.py</code>
limit $\lim$	Richardson on $h=1/n$ (A8 plateau)	App E
ODE $y'=f$	$RK4 = I_\varepsilon$ of the field	App E
series / special functions	Euler–Maclaurin / acceleration	App D/E

classical operation	Λ -preimage (finite)	witness
divergent-series value	regularization = finite retained readout (Part IX)	App D

The exact rung is machine-checked (FTCC, the D_ε rules); the numeric rung is **100/100** (App E) plus **6/6 dimensional cases up to 11-D + non-integer d** (§20.6; 18/18 in that suite overall). The classical object is **never formed** — only its readout is.

20.6 Multi-dimensional integration and dimensional regularization as finite readouts

The dimensions physics actually uses (3-D space, 3+1=4-D spacetime, d -dimensional loop integrals, 10/11-D of string/M-theory) are all covered by one theorem.

Th 20.2 (the d -dimensional Gaussian is a finite readout, for every d). For any dimension d — integer or non-integer — $\int_{\mathbb{R}^d} e^{-|x|^2} d^d x = \pi^{d/2}$, reproduced by the **finite radial readout** $\int_0^\infty (2\pi^{d/2}/\Gamma(d/2)) \cdot r^{d-1} \cdot e^{-r^2} dr$ (the surface factor $2\pi^{d/2}/\Gamma(d/2)$ carries all d dimensions). No completed infinity, no infinite lattice. *Validated (finite_diagnostic, ≥ 10 digits, RAM-light): $d=3$ (space), $d=4$ (spacetime), $d=7$, $d=11$ (M-theory), and $d=4-\varepsilon$ (dimensional regularization), plus a genuine 3-D lattice grid — **6/6 dimensional cases** (18/18 in that suite overall; the other 12 rows are 1-D classical-infinite checks), [validation/infinity_accuracy.py](#).*

- **Cor 20.2a (dimensional regularization is a readout, not magic, D_r).** QFT's dimensional

regularization — evaluating a loop integral at $d=4-\varepsilon$ — is the maya at a declared *non-integer* dimension: $\pi^{(4-\varepsilon)/2}$ is a finite Γ -readout defined for any complex d . The regulator ε is a readout parameter; the $\varepsilon \rightarrow 0$ renormalized limit is the declared $+\mathbb{R}$ -Open step, taken only after A8 stability. The framework does not *inherit* dim-reg from physics — the Gaussian/ Γ object here is a **representative instance** of dim-reg's finite content (same radial-integral + Γ -in- d mechanism); the general propagator-power loop $\int d^d k / (k^2 + \Delta)^n$ reduces to the same Γ -readout but is not computed here.

20.7 Worked examples, and the completeness verdict

- **4-D spacetime:** $\int_{\mathbb{R}^4} e^{-r^2} d^4 x = \pi^2$ — radial readout $\int_0^\infty 2\pi^2 r^3 e^{-r^2} dr$, plateau π^2 to 12

digits at $N=6000$.

- **11-D (M-theory):** $\int_{\mathbb{R}^{11}} e^{-r^2} d^{11} x = \pi^{11/2}$ — same radial readout at $d=11$, 50 digits.
- **dim-reg:** $\int_{\mathbb{R}^{\{3.9\}}} e^{-r^2} = \pi^{\{1.95\}}$ — the identical readout at non-integer $d=4-\varepsilon$, 12 digits.

Verdict — the continuum-maya is complete as a computational replacement (Dr, witnessed). Across the standard toolkit and across the dimensions physics uses, the maya reproduces every classically-continuum value from a finite discrete readout — 100/100 (1-D toolkit) + 6/6 multi-D dimensional cases up to 11-D and non-integer d) + the exact machine-checked core. **Anything the continuum computes, the maya computes finitely; the completed continuum is dispensable for computation.** (*Honest fence, unchanged: this is computational replacement, not a claim to decide the continuum's existence/uniqueness questions — those stay $+\mathbb{R}$ -Open*; declining to form the completed object is the whole point, §0.3, §10.13.)

Part XXI — The frontier without the continuum: paradox dissolution (a decisive stance)

Topology, differential geometry (manifolds), and PDE were listed $+\mathbb{R}$ -Open (§10.13). This Part states the **decisive position**: that mark is **not a computational gap** — it is the *deliberate refusal to reify a non-readout*. **The actual continuum is not needed for computing any of these** — the continuum-maya (Part XX, finite- ε) **computes them all, robustly**; for *stating and deciding* the classical completed-continuum existence/uniqueness/cardinality questions themselves we take no position and form no such object (they stay $+\mathbb{R}$ -Open **by choice**). Every classical *paradox* in these areas is an **artifact of injecting a non-readout into the computation**, dissolved the moment the injection is refused — the classical question about the *object* is not thereby answered, only declined.

21.1 Stance: computation is fully covered, the continuum is optional

- **Topology** — the practical invariants are already discrete/combinatorial: connectivity $= \ker L_R$

(§15.2), Euler characteristic $\chi = V - E + F$ (§15.3), homology of a simplicial complex (finite linear algebra over $\mathbb{Q}$). No completed point-set continuum is needed to compute them.

- **Manifolds / differential geometry** — curvature is the **second retained difference** with a

discrete Gauss–Bonnet (§10.10, DEC toolkit): the vertex angle defects sum to $2\pi\chi$. The **π -free content** is combinatorial and Th_coqc -elig — *the total defect is a topological invariant fixed by χ* ; writing it with 2π and radian incident angles is the **continuum readout** (route (ii) of §4.4, $+\mathbb{R}$ -Open), a `finite_diagnostic` value at declared precision, not an exact $\mathbb{Q}$ readout.

- **PDE** — every PDE is *actually computed* by discretization (not "solved" in the classical existence/uniqueness sense); the finite- ε schemes of Part XI (D_ε , I_ε , the difference-equation ODE, §11.4) **are** the computation. The "continuum limit $L_{R \rightarrow \square}$ " (§10.13) is the *appearance*; the computation lives at finite ε .

So the stance is decisive: we do not need the actual continuum to compute topology, geometry, or PDE — the maya suffices, robustly, at every case that has an answer. What stays $+\mathbb{R}$ -Open is only the *completed continuum object itself*, which we **decline to form** — and that declination is precisely where the paradoxes are dissolved.

21.2 Paradox dissolution — each paradox is an injected non-readout

Every famous paradox of the continuum arises from injecting a **non-readout** (I1–I4 infinities / Z1–Z4 zeros, Part 0; the Axiom-of-Choice over an uncountable domain). Refuse the injection and the paradox does not arise; the maya still returns the practical value. (Diagnoses are *Dr*; the computed values are *finite_diagnostic*.)

paradox (area)	injected non-readout	readout-first dissolution	the maya still computes
Zeno (motion/divisibility)	I2 infinite subdivision $h \rightarrow 0$	finite τ_c floor (Th 2.9); motion = finite discrete steps	I_ε sums the finitely-many steps exactly
Banach–Tarski (topology/measure)	uncountable AC + non-measurable sets	a non-measurable set has no admissible G_λ description — it is not a readout; AC-over-uncountable is refused	μ_λ is always finitely additive; no readout is ever duplicated
Continuum Hypothesis (set theory)	the completed uncountable $\aleph$	$\mathbb{R}$ $\aleph +$ intermediate cardinals	$\aleph$ $\mathbb{R}$ $\aleph$ as a completed cardinal is a non-readout; only finite/countable retained counts exist (§10.2)
Weierstrass (nowhere-differentiable)	completed $h \rightarrow 0$ limit	at declared λ every readout function has a finite D_ε ; "nowhere differentiable" is an $h \rightarrow 0$ artifact	D_ε returns the slope at the declared resolution

paradox (area)	injected non-readout	readout-first dissolution	the maya still computes
Gabriel's Horn (finite volume, infinite area)	I3/I4 reached $+\infty$	both quantities are finite readouts at the declared cutoff ; $+\infty$ is never reached	I_ϵ computes both to the cutoff — no paradox
Koch / fractal (infinite perimeter)	I2 infinite iteration	finite iteration count; perimeter is finite at declared resolution	I_ϵ at resolution λ gives the finite perimeter
Thomson's lamp / supertask	completed infinite sequence	no completed infinite process (Th 10.4–10.5); the "final state" is a non-readout	the state after any <i>finite</i> stage is a plain readout
Russell's paradox (foundations)	unrestricted comprehension (set of all sets)	sets are G_λ -fibers (admissible, finite, no self-membership, §10.1); the paradoxical object is a non-readout	set operations on admissible fibers are decidable
Skolem (countable model of "uncountable")	absolute "uncountable"	"uncountable" is readout-relative , not an absolute property; no contradiction at the readout level	—

paradox (area)	injected non-readout	readout-first dissolution	the maya still computes
Navier–Stokes blow-up / non-uniqueness (PDE)	the completed continuum solution	the blow-up lives in the $+\mathbb{R}$ -Open completed limit; the finite- ϵ scheme (run within its stability bound) returns a finite value at declared λ — the classical existence question is declined, not decided	the difference scheme (§11.4) computes the flow at declared λ

The decisive reading. These are not open problems the framework *fails*; they are non-readouts the framework *declines to inject*. The paradox is the symptom of the injection; refusing the injection is the cure. The continuum-maya (Part XX) then supplies every value the working mathematician actually needs — so the continuum is, for computation, **optional**. (*Honesty fence: this dissolves the paradoxes and computes the readouts; it does not claim to have decided the classical questions in their own completed-continuum terms — declining to form that object is the whole point, §0.3, §10.13.*)

21.3 The three frontier areas, made explicit — the paradox, and the computation that never hits it

(Each is demonstrated numerically in `validation/paradox_dissolution.py`. Diagnoses `Dr`; computed values `finite_diagnostic`; the discrete Gauss–Bonnet identity's π -free/combinatorial content is `Th_coqc`-elig, its 2π /radian-angle form is a $+\mathbb{R}$ -Open continuum readout — §4.4.)

Topology — the paradox: Banach–Tarski (a solid ball cut into finitely many pieces and

reassembled into *two* identical balls; measure is created from nothing).

- **Where it comes from:** the pieces are **non-measurable sets**, built with the Axiom of Choice over

an *uncountable* index — pure non-readouts (`I1` + uncountable AC). Volume is "created" only because the pieces have no admissible volume readout to begin with.

- **Why our computation never hits it:** a readout region is a G_λ -fiber with a retained count

$\mu_\lambda = I_\varepsilon(1_S)$; μ_λ is finitely additive by construction (§16.1), so a finite decomposition and reassembly **conserves the count exactly** — you cannot read out 1 piece as 2. The non-measurable "pieces" simply have no G_λ description, so they are never formed. *Computed:* split any finite region into disjoint parts and recombine — $\mu_\lambda(\text{whole}) = \sum \mu_\lambda(\text{parts})$ on the nose, every time (no doubling possible).

Manifolds / differential geometry — the paradox: curvature requires a smooth limit (the

"how can a *polyhedron* have curvature — it's flat faces and sharp corners?" tension, and the dependence of classical curvature on an $h \rightarrow 0$ that never terminates).

- **Where it comes from:** injecting the smooth chart and $h \rightarrow 0$ (I2) to define curvature pointwise on

a continuum manifold.

- **Why our computation never hits it:** curvature is the **angle defect** concentrated at a vertex —

a finite retained second difference — and **discrete Gauss–Bonnet** ties it to topology: $\sum_{\text{vertices}} (2\pi - \sum \text{incident angles}) = 2\pi \cdot \chi$ (§10.10). No chart, no $h \rightarrow 0$. *Computed (finite_diagnostic — the $2\pi/\text{radian-angle}$ form is the continuum readout, route (ii) of §4.4, $+\mathbb{R}$ -Open):* for a tetrahedron the vertex defects sum to $4\pi = 2\pi \cdot 2$ ($\chi = 2$); for a cube likewise $2\pi \cdot 2$. The **exact, π -free** statement is combinatorial: the total defect is a topological invariant fixed by χ , with the surface built of flat faces.

PDE — the paradox: Navier–Stokes blow-up / ill-posedness (does a smooth solution exist for all

time, or can it become singular — infinite velocity — in finite time?).

- **Where it comes from:** the question is posed in the **completed continuum**; the "blow-up" is a

reached $+\infty$ (I3) inside an idealized $\mathbb{R}^3 \times \mathbb{R}$ that is itself a non-readout.

- **Why our computation never hits it:** a PDE is a difference equation (§11.4); the finite- ε

scheme $y[n+1] = y[n] + I_\varepsilon(\text{field})$, **run at a resolution satisfying the scheme's own declared stability bound** (A8 — e.g. $r = \alpha \Delta t / \Delta x^2 \leq 1/2$ for the explicit heat stencil), returns a finite readout at every step *inside that stability window* — there is no $+\infty$ to reach. (*Outside the window the explicit scheme itself diverges numerically; that is a discretization-stability failure, orthogonal to and not evidence of a continuum blow-up.*) *Computed (finite_diagnostic, at a stable r):* the LINEAR MODEL PDEs — heat $u_t = u_{xx}$

(explicit stencil, $r=0.25$) decaying a bump to its declared floor, and a transport step advancing at finite speed — return finite fields for all n , no blow-up. **Navier–Stokes itself (nonlinear, vortex-stretching) is not run here**; its completed-continuum existence/uniqueness question stays $+\mathbb{R}$ -Open **by choice** — a distinct, declined question about a non-readout — while the finite- ε value the engineer needs is computed without paradox.

The mathematical bounds we have discovered (the PDE frontier, made precise). The stance is not hand-waving — the retained-information PDE layer carries the following bounds (*the relaxation-inertia equation and its energy identity are from the URCF turbulence pack; the telegraph coarse-grain is the PGFT/DHRG substrate form; the spectral bound is the standard Davis–Kahan theorem applied to L_R here — Lahtee; tiers as marked*):

- **Relaxation-inertia equation.** The retained turbulence/field state obeys

$\tau_R \cdot dI_R/dt + L_R \cdot I_R = S_R + \eta_R$ — a first-order relaxation under the retained operator L_R , inertia $\tau_R > 0$, drive S_R , and a **logged, bounded** residual η_R (D_r ; η_R is not a free error bin). Its mild solution is the semigroup $I_R(t) = e^{\{-t \tau_R^{-1} L_R\}} I_R(0) + \dots$.

- **No-blow-up bound.** For the homogeneous relaxation $\tau_R dI_R/dt = -L_R I_R$, the standard energy

identity gives $d/dt \|I_R\|^2 = -(2/\tau_R) \cdot \langle I_R, L_R I_R \rangle = -(2/\tau_R) \cdot B(I_R, I_R)$ (D_r — the ODE/derivative step, not machine-checked here). The **decisive sign** is machine-checked: $B(I_R, I_R) \geq 0$ and hence $-(2/\tau_R) \cdot B(I_R, I_R) \leq 0$ is `Th_coqc` (witness `formal/IDM_Keystone.v: relaxation_dissipation`, resting on the keystone $L_R \geq 0$, §5.1). So a positive-semidefinite retained operator **cannot grow the homogeneous mode** — the rigorous content behind "finite- ε schemes don't explode" is the machine-checked positive-semidefiniteness of L_R , not luck; the dissipation *identity* itself is the standard D_r step.

- **Finite propagation speed (D_r).** The telegraph coarse-grain $\tau_c \cdot u_{tt} + u_t = D \cdot \Delta u$ (the PGFT/DHRG

substrate form) is a telegrapher's equation, which propagates signals at the **finite speed** $\sqrt{(D/\tau_c)}$ (standard, not sourced verbatim) — no infinite propagation (unlike the parabolic heat limit $\tau_c \rightarrow 0$). The $+\infty$ -velocity pathology is an artifact of the $\tau_c \rightarrow 0$ non-readout.

- **Spectral-mode stability, Davis–Kahan (D_r).** By the standard Davis–Kahan $\sin \theta$ theorem applied

directly to L_R , its dominant retained mode (the Fiedler mode) is recovered under an operator perturbation E with $\sin \theta \leq 2\|E\| / \Delta_{\text{spec}}$, where Δ_{spec} is **L_R 's own spectral gap** — the mode is stable exactly when that gap dominates the noise. (*Caveat,*

per the DHRG source: this dynamical gap Δ_{spec} of L_R is distinct from and must not be conflated with the data-recoverability gap σ_a^2 in DHRG's readout-side Theorem 5, whose bound is $\sin \theta \leq 2\|N\|/\sigma_a^2$; both gaps are separately required.)

(Honesty fence, inherited from the source packs: these are bounds on the **retained-information structural model** — finite-dimensional, over our L_R — and a theorem-facing scaffold; they are **not** a Navier–Stokes continuum closure. The completed-continuum existence/regularity theorem stays $+\mathbb{R}$ -Open, its own declared target — §10.13.)

Summary of the stance for the three areas: the paradox is always in the *completed-continuum object* (non-measurable set · smooth-limit curvature · reached $+\infty$), never in the *computation*. Refuse the object, keep the computation: topology by μ_λ / L_R , geometry by discrete Gauss–Bonnet, PDE by finite- ε schemes — **all three computed robustly, none producing its paradox.**

Part XXI.5 — The Hilbert-space mathematical core (finite-dim exact, $+\mathbb{R}$ -fenced)

A **Hilbert space is pure mathematics** — linear algebra / functional analysis / operator theory — *not* physics. This Part builds it as an operator-theoretic **information architecture**, provable entirely inside mathematics; the physical interpretation (§8) is a detachable Dr layer that no theorem here needs. Witnesses: `idm/hilbert.py` (+ `idm/hilbert_open.py` for the frontier), `formal/IDM_Hilbert.v` (13 theorems, axiom-free), `HILBERT_PLUS_R_FRONTIER.md`.

21.5.1 The thesis

The mathematical core is `Core = (H, ⟨·,·⟩,  $\mathcal{O}$ ,  $\mathcal{T}$ )` — the space, its operator algebra, and the information-transformation rules. We claim only the mathematical level: organize, reduce, and compute information over a complete inner-product space. No particle/wave/measurement/quantum claim is made.

21.5.2 The construction tower (each arrow tier-tagged)

`Set → Field → Vector Space → Inner-Product Space → Normed Space → (Complete Space) → Hilbert Space`. Our build climbs to the **finite-dimensional exact** rung and stops: `(Complete Space)` is the I1 ($\mathbb{R}$ -completeness) rung, boxed as $+\mathbb{R}$ — entered only as a readout (§21.5.7).

21.5.3 The operator ladder (finite $n \times n$ throughout)

`Linear → Bounded → Adjoint → Self-adjoint → Unitary → Normal → Spectral theory`.

21.5.4 The finite-dim exact core

Over $\mathbb{Q}$ or the Gaussian rationals $\mathbb{Q}[i]$ (an exact QC pair — **never** IEEE complex), finite n :

- **inner product** `⟨u,v⟩ = Σ conj(ui)vi`, exact; sesquilinear + PSD — machine-checked (`inner_sym` [Toledo: Keystone/P.04.v1],

`inner_linear_l` [Toledo: Keystone/P.05.v1], `inner_pos` [Toledo: Keystone/P.06.v1]). **Cauchy–Schwarz** (`cauchy_schwarz_2` [Toledo: Keystone/P.09.v1]), **parallelogram law**, **Pythagoras** (`pythagoras_orthogonal` [Toledo: Keystone/P.08.v1]) — all axiom-free $\mathbb{Q}$.

- **Gram–Schmidt**, split as *orthogonalize* (exact $\mathbb{Q}[i]$, field division only — no $\sqrt{}$) then *normalize*.

Normalization is exact iff every squared norm is a perfect square in $\mathbb{Q}$, else `finite_diagnostic` (a certified numeric $\sqrt{}$). This per-instance split is the honesty core, not a blanket " $n \leq k \Rightarrow \text{exact}$ ".

- **projection** $P=A(A^*A)^{-1}A^*$ (exact), idempotent + self-adjoint (`projection_idempotent` [Toledo: Keystone/P.14.v1],

`projection_self_adjoint` [Toledo: Keystone/P.15.v1]). **adjoint / self-adjoint / unitary / normal** — decidable exact $\mathbb{Q}[i]$ equalities; $(A^*)^* = A$ and $(AB)^* = B^*A^*$ machine-checked (`adjoint_involutive` [Toledo: Keystone/P.10.v1], `adjoint_of_product` [Toledo: Keystone/P.11.v1]).

- **characteristic polynomial** — exact $\mathbb{Q}[i]$ coefficients (Faddeev–LeVerrier), always exact regardless

of root rationality. **spectral decomposition** of a Hermitian/normal matrix: **exact only if every eigenvalue is found rational** (via `characteristic_poly` + `rational_roots`) and eigenvectors normalize exactly; **otherwise a certified numeric eigensolver with a proven residual $\|MV-VA\| \leq \varepsilon$, tier `finite_diagnostic`** — decided per call, never a general- n "real spectrum" claim. The Hermitian 2×2 reality-of-spectrum content is machine-checked, as a fixed instance: `hermitian_2x2_discriminant_nonneg` [Toledo: Keystone/P.12.v1] ($0 \leq (a-c)^2 + 4b^2$) and `hermitian_2x2_gap_is_discriminant` [Toledo: Keystone/P.13.v1] ($(\lambda_1 - \lambda_2)^2 = \text{discriminant}$, Vieta). **operator norm** is *always* `finite_diagnostic` (largest singular value, certified residual); a separate exact sqrt-free `gershgorin_bound` gives an operator-norm *upper bound*, never the true sup norm.

21.5.5 Composite systems (operator algebra, physics-neutral names)

tensor product (Kronecker, exact $\mathbb{Q}[i]$), **partial trace** (exact block sum), **Choi matrix** of a finite completely-positive-map generator list, **complete positivity** = the Choi matrix is PSD — tested by the **eigenvalue sign**, *not* Sylvester leading principal minors (those test positive-definiteness; `diag(0,-1)` passes the leading minors yet is not PSD — a worked example of tier discipline catching a wrong proof).

21.5.6 Separability as an open-in-general decision problem

The **Peres–Horodecki (PPT)** test: a *negative* partial-transpose eigenvalue is an **exact witness of entanglement** (any dimensions); PPT is necessary-and-sufficient only in $2 \times 2 / 2 \times 3$ (`finite_diagnostic` "separable" there); outside that range a positive partial transpose is inconclusive and returns **HOLD**, never "separable" — this domain's own instance of *diagnose, don't overclaim*.

21.5.7 The $+\mathbb{R}$ fence (completeness / infinite dimension)

Completeness, ℓ^2 , $L^2(X, \mu)$, infinite orthonormal bases, and the infinite-dim spectral theorem are $+\mathbb{R}$ -Open: **never formed**, only named readouts of finite $\mathbb{Q}$ -approximants

(`idm/hilbert_open.py`). The fence is **code-enforced** — `hilbert.py` never imports `hilbert_open.py` (AST-tested), and the open readouts return `status:"+R_OPEN"` with no `value` field, so `solve()` cannot emit an infinite-dim result as certified. Full statement: [HILBERT_PLUS_R_FRONTIER.md](#) .

21.5.8 Physics → math dictionary (Dr , detachable, boxed)

Optional and removable; no theorem in §21.5.4–§21.5.6 depends on it:

physics language	pure-math object	physics language	pure-math object
quantum state	unit vector in H	superposition	linear combination
observable	self-adjoint operator	entanglement	non-separable tensor element
evolution	unitary $x_{t+1}=Ux_t$, $U^*U=I$	gate	unitary operator
measurement	projection / spectral decomposition	channel	completely-positive linear map
amplitude	(complex) coefficient	collapse	projection / conditional update

21.5.9 What this Part is NOT

No quantum-computer, quantum-simulation, or empirical-physics claim. §21.5.8 is not evidence for the theorems, and the theorems do not require §21.5.8 to be meaningful. Naming: *Hilbert-space computational architecture / operator-theoretic information architecture* — never "quantum computer / simulates nature."

Part XXII — The root design: three axioms that generate the solver

The preceding parts build mathematics rung by rung. This part states the **minimal generating core** — the shortest set of axioms from which the book's *computational* content, and the branch half of the solver `idm` (the 230-kind branch map; 258 with the Part XXI.5 Hilbert core, 259 live including the `polynomial_positivity` certificate solver), is produced. The claim of this part is deliberately narrow and tier-honest: **three axioms plus one bridge, of which three have machine-checked cores, and whose branch reductions are machine-checked in `formal/IDM_Reduction.v` (axiom-free).**

22.1 The core — { `READOUT` · `FOLD` · `DECISION` } + the KEYSTONE bridge

- **A1 · `READOUT`** (`Dr`, the substrate — Part 0.5, Part II).

Every appearance is a finite retained difference `δ_R`, a readout `r = 0_ε(X) ∈ ℚ` at a declared resolution `ε > 0`. `ℝ`, `+∞`, and the point of zero extent are **non-readouts** — boundaries, never appearances. *This is the framing that makes the rest finite; it is a stance, not a theorem.*

- **A2 · `FOLD`** [Toledo: A2] (`Th_coqc`, the engine — Part VIII, §22.3).

One generic accumulation over a carrier and operation,

$$I_{\oplus}[f](N) = \bigoplus_{k < N} f[k], \quad I(Df) = f[N] \boxminus f[0].$$

Changing `(⊕, ⊗)` changes the branch of mathematics; the telescoping/FTCC identity is exact.

- **A3 · `DECISION`** [Toledo: A3] (`Th_coqc`, the search — §22.3).

Some problems are not accumulations but existence questions:

$$P(X) \iff \exists w \text{ finite} : \text{check}(X, w) = \top \quad (\text{no witness} \Rightarrow \mathbf{HOLD}).$$

This is also the honesty mechanism: absence of a witness is a refusal, never a guess.

- **A4 · `KEYSTONE`** [Toledo: Keystone] (`Th_coqc`, the bridge — Part V, Th 5.1).

$$B(\Phi, \Phi) = \sum_e w_e (\Phi_i - \Phi_j)^2 = I(\Phi), \quad L_R = D_W - W.$$

Because `D_W` is the very difference operator of A2, the keystone turns the forward FOLD engine into a **variational / inverse** engine (optimization, physics, inverse problems, the RCP savings) — this is what makes the core "hard" enough for real-world engineering, not only evaluation.

22.2 Why this is the shortest maximal core

Three modes are genuinely distinct and none is redundant: **accumulate** (A2), **search-and-certify** (A3), and **the substrate they run on** (A1); A4 is the one bridge that lifts computation to optimization. Everything else in the solver is an *instantiation* — a choice of semiring $(\oplus, \otimes)$ for a FOLD, or a choice of `check` for a DECISION.

22.3 The reductions, machine-checked (`formal/IDM_Reduction.v` , `Th_coqc` , axiom-free)

The structural claim "each branch's kernel is an instance of A2 or A3" is **not left as narrative** — it is proven, Closed under the global context :

theorem (witness <code>IDM_Reduction.v</code>)	reduces which kernel
<code>ftcc_Z</code> [Toledo: A2/M.03.v1]	analysis / summation branch = fold over $(+)$; telescoping $I(Df)=f[N]-f[0]$ exact
<code>foldmin_le_init</code> [Toledo: A2/M.04.v1], <code>foldmin_le_elem</code> [Toledo: A2/M.05.v1]	shortest-path / DP branch = the same fold over $(\min)$ (relaxation invariant)
<code>foldmax_ge_init</code> [Toledo: A2/M.12.v1], <code>foldmax_ge_elem</code> [Toledo: A2/M.13.v1]	critical- / widest-path branch = the same fold over $(\max)$
<code>sum_is_fold</code> [Toledo: A2/M.06.v1], <code>path_is_fold</code> [Toledo: A2/M.07.v1]	the two kernels are definitionally one fold , differing only in $(\oplus, \otimes)$
<code>fold_linear</code> [Toledo: A2/M.10.v1], <code>fold_add_split</code> [Toledo: A2/M.11.v1]	the semiring laws (linearity, additivity) that <i>make</i> the reduction valid
<code>dot_is_fold</code> [Toledo: A2/M.17.v1], <code>dot_scale</code> [Toledo: A2/M.18.v1]	inner product / matrix-multiply / convolution kernel = fold of products
<code>pivot_preserves</code> [Toledo: A2/M.08.v1]	linear-algebra / LP / determinant-sign geometry atom: a retained-difference row op preserves every solution $\Rightarrow$ elimination is a schedule of folds
<code>horner_scaled</code> [Toledo: A2/M.19.v1], <code>horner_is_poly</code> [Toledo: A2/M.20.v1]	polynomial branch: Horner's method is a (right) fold and computes the polynomial $\sum c_k x^k$ exactly
<code>foldmin_in</code> [Toledo: A2/M.21.v1]	shortest-path exactness: the accumulated minimum is attained (equals the start or an actual element), not merely a lower bound
<code>fold_add_app</code> [Toledo: A2/M.22.v1]	the associative-monoid chunking law: a fold splits over concatenation $\Rightarrow$ associative reductions parallelise (MapReduce), order-independent
<code>factorial_is_fold</code> [Toledo: A2/M.23.v1]	product / combinatorics branch: $n!$ is the same fold over $(\times)$

theorem (witness <code>IDM_Reduction.v</code>)	reduces which kernel
<code>witness_sound</code> [Toledo: A3/M.01.v1], <code>witness_complete</code> [Toledo: A3/M.02.v1], <code>decide_reflect</code> [Toledo: A3/M.06.v1], <code>decide_dec</code> [Toledo: A3/M.03.v1]	A3: a bounded checkable witness is a decidable ACCEPT/HOLD, faithfully reflected
<code>witness_composite_sound</code> [Toledo: A3/M.04.v1], <code>composite_has_factor</code> [Toledo: A3/M.05.v1]	number-theory kernel: a found divisor proves compositeness (the <code>is_prime</code> backbone)
<code>witness_power_sound</code> [Toledo: A3/M.07.v1]	<code>integer_root</code> / <code>is_perfect_square</code> kernel: a found <code>w</code> with <code>w^k=n</code> proves a perfect power
<code>witness_qr_sound</code> [Toledo: A3/M.08.v1], <code>witness_dlog_sound</code> [Toledo: A3/M.09.v1]	crypto kernels: a found <code>w</code> with <code>w^2≡a / g^w≡h (mod p)</code> solves the modular square root / discrete logarithm
<code>sat_reduces_to_decision</code> [Toledo: A3/M.10.v1], <code>sat_model_sound</code> [Toledo: A3/M.11.v1]	the flagship decision problem: CNF satisfiability over the 2^n assignments is the A3 schema; a returned model provably satisfies every clause
<code>relax_nonincreasing</code> [Toledo: A2/M.24.v1], <code>relax_idempotent</code> [Toledo: A2/M.25.v1]	Bellman–Ford / DP convergence : min-plus relaxation is non-increasing and idempotent per edge $\Rightarrow$ the fixpoint is reached in finite rounds
<code>weak_duality_2</code> [Toledo: A2/M.26.v1]	A2 $\times$ A3 bridge — LP weak duality: a dual-feasible $y \geq 0$ is a <i>certificate</i> (A3) that bounds the simplex objective (A2), $c \cdot x \leq y \cdot b$
<code>fold_right_perm</code> [Toledo: A2/M.02.v1] (+ <code>op_swap</code> [Toledo: A2/M.01.v1])	the deep semiring law : a fold over an associative- commutative operation is invariant under any reordering (Permutation) of its inputs — the licence to reorder / chunk / parallelise, so A2 is well-defined independent of traversal order
<code>fold_split_even_odd</code> [Toledo: A2/M.27.v1]	the FFT divide-and-conquer split: the sum over $[0, 2n)$ decomposes into the even-index and odd- index sub-folds (the recursive decomposition the fast transform is built on)
<code>modpow_is_fold</code> [Toledo: A2/M.28.v1]	crypto forward map: modular exponentiation $g^N \bmod p$ is the same fold over $(\times \bmod p)$ — the RSA / EC / discrete-log engine
<code>foldmax_in</code> [Toledo: A2/M.14.v1]	critical-/widest-path exactness: the accumulated maximum is attained (mirror of <code>foldmin_in</code> [Toledo: A2/M.21.v1])
<code>sum_list_perm</code> [Toledo: A2/M.15.v1], <code>prod_list_perm</code> [Toledo: A2/M.16.v1]	set-aggregation is order-independent : Σ and Π over a multiset are well-defined (corollaries of <code>fold_right_perm</code> [Toledo: A2/M.02.v1]) — the basis of <code>describe</code> /statistics and combinatorial products
<code>tautology_sound</code> [Toledo: A3/M.12.v1], <code>cnf_tautology_sound</code> [Toledo: A3/M.13.v1]	the universal counterpart of the witness search — <code>truth_table</code> / validity: a bounded $\forall$ over the 2^n rows; a <code>true</code> verdict proves the formula on every row

theorem (witness <code>IDM_Reduction.v</code>)	reduces which kernel
<code>witness_crt_sound</code> [Toledo: A3/M.14.v1]	the <code>crt</code> kernel: a witness <code>x</code> meeting every congruence <code>x ≡ r_i (mod m_i)</code> solves the simultaneous system

22.4 The branch map — 230 solver kinds, exactly partitioned

*This is the A2/A3 **branch map** (230 kinds). The Hilbert-space core (Part XXI.5) adds a further 28 kinds — inner products, projections, Gram/Gram–Schmidt, operator norms, adjoint/self-adjoint tests, spectral decomposition — for 258; with the `polynomial_positivity` certificate solver the **live registry total is 259** (`idm.kinds()`). The partition below is the fold-generated branch half; it is not weakened by the Hilbert extension sitting beside it.*

branch	generated by	count	representative kinds
Analysis / continuum frontier	A2 fold over <code>(+, ×)</code>	97	<code>integral</code> , <code>ode</code> , <code>pde_*</code> , <code>taylor_series</code> , special functions, transforms
Paths / dynamic programming	A2 fold over a tropical semiring	22	<code>shortest_path</code> , <code>widest_path</code> , <code>knapsack</code> , <code>mst</code> , <code>max_flow</code>
Linear algebra / polynomials / stats	A2 fold with pivot (\$ <code>pivot_preserves</code> [Toledo: A2/M.08.v1])	46	<code>matrix_*</code> , <code>eigenvalues</code> , <code>linear_program</code> , <code>regression</code>
Number theory / crypto / logic	A3 finite witness	55	<code>factorize</code> , <code>is_prime</code> , <code>sat</code> , <code>discrete_log</code> , <code>rsa_*</code>
Geometry	A2 sign of a determinant (a signed fold)	10	<code>orient</code> , <code>convex_hull</code> , <code>in_circle</code> , <code>polygon_area</code>
total		230	

`97 + 22 + 46 + 55 + 10 = 230` (branch partition verified against the pre-Hilbert registry; with the 28-kind Hilbert-space core of Part XXI.5 that is 258, and the `polynomial_positivity` certificate solver brings the live registry to 259).

22.5 Honest fence (declared, not hidden)

- **Proven (`Th_coqc`).** The branch **kernels** reduce to A2/A3 (the theorems of §22.3), axiom-free; the

keystone A4 and the additive FTCC are machine-checked (Part V, Part VIII).

- **Not proven.** That every one of the 230 *Python implementations* matches its kernel — the reductions

certify the mathematical form, not each line of code. A1 (READOUT) is a **stance**, not a theorem. The full statement "A2+A3 generate all 230 kinds" is `Dr` (design); §22.3 lifts the *kernel* half of it to `Th_coqc`, and further coverage is the honest open work.

- **Not given for free.** Transcendental *values* (π , e) are readout-invariants that *emerge* from

the fold, not derived from the axioms; and *modeling* (choosing `f`, choosing the semiring for a real problem) is human input — the core is the solver, not the modeller.

Appendix A — The contaminated-concept → discrete-replacement table

contaminated concept	injection	discrete-correct replacement
real number $\mathbb{R}$ / completeness	I1	$\mathbb{R}$ = readout (regular Cauchy of $\mathbb{Q}$); $\sqrt{2}$, π non-readouts (InfoIrrationalNonReadout)
the point ($r=0$)	Z1	a node / retained distinction (graph vertex)
zero as occupied	Z1/Z3	refused non-readout, or the L_R kernel = indistinguishability (not a void)
$+\infty$ / $N \rightarrow \infty$ / limit that lands	I3/I4	$\mathbb{Q}$ has no $+\infty$; a limit is the finite approach, never the endpoint
infinite divisibility $h \rightarrow 0$	I2	finite step / τ_c floor (Th 2.9)
angle / degree	I1 (inverse trig)	overlap fraction (Def 4.2a) / turning number (Def 4.2b)
continuity / smooth (ε - δ over $\mathbb{R}$)	I1+I2	discrete Lipschitz; ε - δ is a derived $+\mathbb{R}$ rung (Th 3.9)
distance = coordinate difference	$\mathbb{R}$ -frame	accumulated retained resistance along optimal path (Def 4.1)
line / continuum as substrate	primitive continuum	continuum = readout of the discrete graph (Pr 3.2)
π , e , ϕ as numbers	transcendental-as-real	readout-invariants (Pi_as_Readout_Invariant)
trichotomy / total $\leq$ / LUB	LPO/WLPO omniscience	cotransitivity + Cauchy-complete + finite lattice (Th 3.7–3.8, Pr 3.1)
derivative / integral = continuum limit	I2	discrete Δ , Σ , discrete FTC (Th 3.2–3.3)
operator on a continuum (∂^2)	I2	graph Laplacian L_R (Th 4.3); $\partial^2 a$ $+\mathbb{R}$ readout

Appendix B — Machine-checked theorem index (tier · witness)

Local witnesses (this repo, `formal/*.v` — Coq 8.20, all axiom-free / *Closed under the global context*): `formal/IDM_Keystone.v`: `keystone_B_eq_I` [Toledo: Keystone/M.03.v1] (Th 5.1, $B(\Phi, \Phi) = I(\Phi)$) + `keystone_nonneg` [Toledo: Keystone/M.07.v1] (L_R PSD) + `relaxation_dissipation` [Toledo: Keystone/M.08.v1] (§21.3 no-blow-up). `formal/IDM_Logic.v`: `finite_satisfaction_dec` [Toledo: D/M.58.v1] (§10.3 Th 10.6, finite model-checking decidable) + `rdl_non_explosion` [Toledo: deltaR/M.04.v1] (§10.4/Part I, paraconsistent countermodel). `formal/IDM_FiniteWitnesses2.v`: `same_set_same_size` [Toledo: D/M.45.v1] (Th 10.3), `tape_count_succ` [Toledo: D/M.43.v1] (Th 10.4), `no_infinite_readout` [Toledo: D/M.42.v1] (Th 10.5), `lagrange_order_div` [Toledo: Z/M.10.v1] (§12.3). `formal/IDM_FiniteWitnesses.v`: `kuratowski_pair_inj` [Toledo: deltaR/M.02.v1] (Th 10.1, §10.1) · `handshake_lemma` [Toledo: LR/M.01.v1] (§15.2) · `pigeonhole` [Toledo: D/M.40.v1] (§15.3) · `finite_yoneda` [Toledo: D/M.39.v1] (§17.2) · `semiring_distrib` [Toledo: D/M.41.v1] (§12.2). `formal/IDM_Geometry.v`: `orient_swap_bc/ab`, `orient_cyclic` [Toledo: LR/M.05.v1], `orient_coincident_*`, `orient_translation` [Toledo: LR/M.09.v1], `orient_scale` [Toledo: LR/M.10.v1], `orient_collinear_mid/aff` (Part IV / §22.4, the exact orientation predicate the geometry solver branches on, 10 theorems). `formal/IDM_Reduction.v` (Part XXII, the root-design reductions, 42 theorems): `ftcc_Z` [Toledo: A2/M.03.v1], `foldmin_le_init/elem`, `foldmax_ge_init/elem`, `sum_is_fold` [Toledo: A2/M.06.v1], `path_is_fold` [Toledo: A2/M.07.v1], `fold_ext` [Toledo: A2/M.09.v1], `fold_linear` [Toledo: A2/M.10.v1], `fold_add_split` [Toledo: A2/M.11.v1], `dot_is_fold` [Toledo: A2/M.17.v1], `dot_scale` [Toledo: A2/M.18.v1], `pivot_preserves` [Toledo: A2/M.08.v1] (A2); `witness_sound/complete`, `decide_reflect` [Toledo: A3/M.06.v1], `decide_dec` [Toledo: A3/M.03.v1], `witness_composite_sound` [Toledo: A3/M.04.v1], `composite_has_factor` [Toledo: A3/M.05.v1], `witness_power_sound` [Toledo: A3/M.07.v1] (A3). `formal/IDM_ReadoutMinimality.v` + `IDM_ResolvedCount.v` (the signed-readout value-set + the resolved 0/1 inertia count), `formal/IDM_EquivariantReadout.v` (P1 necessary condition), `formal/IDM_Apriori.v` (the Richardson a-priori stability certificate), `formal/IDM_SetsFunctions.v` (Th 10.2, the function $\equiv$ functional-relation coincidence), `formal/IDM_FirstOrder.v` (Th 10.6 first-order $\models_\lambda$, decidable finite-domain satisfaction). One-command audit: `bash formal/verify.sh` (**174 theorems, all Closed under the global context**). Reproduce a single one: `cd formal && coqc -q IDM_Reduction.v` then `Print Assumptions ftcc_Z`.

Sister-repo witnesses (`solver-arc-private/formal/*.v`, NOT local to this repo — verified directly via `coqc -q RD.v` + `Print Assumptions <name>`): `RDL.v` (RDL logic, 8 thm, `Th_coqc`) · `RD.v` + `RDL_Distinguishability.v` (D semiring/order/PA/ $\text{Con_PA} / D \equiv \mathbb{N} / D \equiv \text{PA} /$ discrete-floor, `Th_coqc`; `Con_PA_classical`

+classic; see §2.3/§10.3 for the $D \models PA / D \equiv \mathbb{N}$ tier note) · number ladder $\mathbb{Z}/\mathbb{Q}/\mathbb{R}$
 (PGFT_Roots_of_Mathematics_and_Geometry corpus, **123 constructive theorems axiom-free**, classical capstone on classic alone) · RDL_CausalOrder.v (lap, B mechanics, Th_coqc) · InfoIrrationalNonReadout, InfoZeroInfinityReciprocal, InfoAgencyZeroNonReadout, InfoOperatorLosesPropertyAtEndpoints (non-readout theorems, Th_coqc) · continuum-analysis rungs (RDL_ContinuumLimit, RDL_TaylorLimit) + $\mathbb{R}$ -axioms.

Appendix C — Axiom-dependence discipline

Author `.v` files with **explicit $\forall$ -premises, never `Section / Hypothesis`** (the source-scan reads those as axioms). A result true only because its premises have no $\mathbb{Q}$ witness is a *schematic conditional* — say so. `coqc` `exit 0` + `Print Assumptions` *Closed* is the only ticket to `Th_coqc`.

Appendix D — Validation: 1000 problems, ประถม → ปริญญาเอก (the release dogfood)

The framework is exercised on **1000 problems across five levels**, each solved by a framework-consistent method (exact $\mathbb{Q}$ operators / discrete causal calculus $D_\varepsilon, I_\varepsilon, \text{FTCC}$ / regularization-residue / asymptotics) and checked against an **independent reference** (fractions, sympy, mpmath). Runnable: `validation/thousand_problems.py`.

level	area	pass
L1 ประถม	arithmetic, fractions, order (๑๑÷)	200/200
L2 มัธยม	$^$, $\sqrt$, quadratics, arith/geom series, gcd, mod, discrete- log	200/200
L3 ปริญญาตรี	FTCC & causal product rule (exact $\mathbb{Q}$), Faulhaber Σi^p , binomial, determinants, Stirling	200/200
L4 ปริญญาโท	$\zeta(2k)$ & γ via Euler–Maclaurin, recurrences, roots of unity, Chebyshev/LLN, Bernoulli	200/200
L5 ปริญญาเอก	$\zeta(-1)=-1/12$, Abel/ η sums, Apéry $\zeta(3)$, Ramanujan $1/\pi$, partition asymptotics, continued fractions, Catalan, saddle-point Stirling	200/200
TOTAL		1000/1000 (100%)

Honest reading (tier-aware). L1–L3 are largely **exact $\mathbb{Q}$** — there the framework's *own* computation is the answer, so these validate the operator and discrete-calculus layers (FTCC and the causal product rule hold with **zero residue**, `Th_coqc`). L4–L5 are **genuine numeric-vs-reference**: finite- ε partial sums + Euler–Maclaurin reproduce $\zeta(2k)$, γ , $\zeta(3)$ to $\sim 1e-8$; regularization gives $\zeta(-1)=-1/12$, $1-1+1-\dots=1/2$ (`finite_diagnostic`, post-A8-stability); Hardy–Ramanujan $p(n)$ and Stirling are **leading-order** (few-%), disclosed as such. An earlier draft showed 2 misses; detailed analysis found the CAUSE was **not the mathematics** but a fragile test-harness step (`sympy.nsimplify`, a float→symbolic *heuristic*, was misused to compare exact large rationals). Replaced by exact $\mathbb{Q}$ equality (`exact_eq`) — the robust long-term fix — after which the suite is **1000/1000**. *(The lesson is itself the framework's: an exact readout must be compared as an exact readout, never through a floating-point guess.) The framework computes the standard mathematics, grade-school to frontier, without ever making the continuum a primitive.*

Appendix E — Validation: 100 world-class CONTINUOUS problems reproduced from the discrete

Purpose. Appendix D showed the framework computes standard mathematics; this appendix discharges the strongest claim — that the **continuum itself is dispensable**. Every problem here is one classical analysis defines through a completed limit (integral, derivative, limit, ODE, special function); each is solved by a **finite- ε discrete method only** (I_ε quadrature + Euler–Maclaurin, D_ε + Richardson, finite- ε limit + Richardson-on- $1/n$, RK4 difference-equation, discrete series/quadrature) and checked against the world benchmark. Suite: `validation/hundred_continuum_problems.py` (deterministic, 30-digit `mpmath` reference; each method is a genuine discrete computation — no continuum solver is called).

block	area	method (discrete)	pass
A	definite & improper integrals ($\int_0^1$, Gaussian, oscillatory, $\int_0^\infty$)	I_ε trapezoid + Euler–Maclaurin; admissible reparametrization	34/34
B	derivatives at a point (polynomial, transcendental, x^x)	D_ε central + Richardson	16/16
C	limits ($(1+1/n)^n \rightarrow e$, $H_n - \ln n \rightarrow \gamma$, $(n!)^{1/n} / n \rightarrow 1/e$, ...)	finite- ε + Richardson on $h=1/n$ (A8 plateau)	15/15
D	ODEs (linear, Riccati, $y' = 2xy = e^{x^2}$, $y' = 1/(1+x^2) = \arctan$)	RK4 = 4th-order I_ε of the vector field	12/12
E	special functions (Γ , $\zeta(s)$, erf , Catalan, Li_2 , elliptic K , Bessel J_0 , Wallis, Dirichlet)	defining discrete series / quadrature / product	23/23
TOTAL			100/100 (100%)

Honest reading (tier-aware). The exact- $\mathbb{Q}$ rungs (FTCC, D_ε rules) carry `Th_coqc`; the numeric rungs are `finite_diagnostic` — the *readout* is reproduced to the declared tolerance ($1e-6$ – $1e-10$ typical; $1e-3$ – $1e-4$ disclosed per-line for genuinely singular integrands), while the completed $\varepsilon \rightarrow 0$ existence stays `+R-open` (§10.13). Two harness pitfalls were caught and fixed for the long term: (1) an endpoint Euler–Maclaurin correction that stepped **outside** $[a, b]$ and returned a complex value — fixed to a one-sided *inward* difference; (2) a benchmark with a **sign error** in a Taylor coefficient ($n^4(\cos(1/n) - 1 + 1/2n^2) \rightarrow +1/24$, not $-1/24$) — the framework had the correct sign; the reference was corrected. **The continuum enters nowhere as a primitive and is recovered everywhere as a readout — the operational meaning of "closing the continuum" (Part XI).**

Roadmap (honest frontier — what this edition states vs. what remains)

Stated & machine-checked: RDL logic; $\delta_{R \rightarrow D \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R}}$ to the ordered-field / Cauchy-complete / lattice level; discrete calculus; metric-space + betweenness geometry; the Laplacian operator mechanics; the non-readout theorems and the endpoint principle; **the operator keystone** $B(\Phi, \Phi) = I(\Phi)$ (Th 5.1, now machine-checked — formal/IDM_Keystone.v). **Deferred / imported (+ $\mathbb{R}$ / Open):** the full continuum derivative-integral tower; continuum geometry (manifolds, curvature), and the physical readouts (spectra $\rightarrow$ masses) built on top.

Information Discrete Mathematics — developed by Yaoharee Lahtee. Readout-not-truth applied to the foundations, from the retained difference up. AI-assisted; the core stance and results are the author's. Tiers are honest; where the continuum enters it is flagged, never smuggled.